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COLLEGE OF COMPUTING AND INFORMATICS

Department of Software Engineering

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AI-Based Smart Library Management and Recommendation System

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APPROVAL SHEET / DECLARATION

The project proposal entitled “AI-Based Smart Library Management and Recommendation System” is our original work and has not been submitted for a degree in any other university. All sources of materials used in this project proposal have been properly acknowledged.

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It is hereby certified that this project proposal has been prepared in accordance with the formatting rules laid down by the College and is found to be adequate in scope and quality for initiating the full project documentation.

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EXECUTIVE SUMMARY

The main objective of this project is to develop an AI-Based Smart Library Management and Recommendation System for Mekdela Amba University Library.

The existing library system is mainly manual or semi-digital, which leads to inefficiencies such as slow book retrieval, difficulty in tracking borrowed materials, and increased workload for librarians and library staff.

To address these challenges, the proposed system automates core library operations including book management, user registration, borrowing and returning processes, reservation handling, digital resource management, and reporting.

The system supports multiple types of users, including Student / Library Users, Academic Staff (Teachers and Researchers), Librarians, System Administrators, and Digital Library Officers, each with specific roles and access rights within the system.

In addition, the system integrates an Artificial Intelligence-based recommendation engine using a content-based filtering approach. The model analyzes user borrowing history, search behavior, and preferences to generate personalized book recommendations that improve resource discovery.

A Digital Library Officer manages electronic resources such as e-books, journals, and digital archives, ensuring efficient access to online academic materials. Meanwhile, the System Administrator oversees system configuration, security, user roles, and overall system performance.

The system is developed using React.js for the frontend, Django for the backend, and MySQL for the database. Machine learning libraries such as Pandas, NumPy, and Scikit-learn are used to implement the AI recommendation module.

The expected outcome is an efficient, user-friendly, and intelligent library system that improves resource accessibility, enhances academic support for students and staff, reduces manual workload, and increases overall library service quality.

LIST OF ACRONYMS

- AI – Artificial Intelligence
- API – Application Programming Interface
- DBMS – Database Management System
- ER – Entity Relationship
- LMS – Library Management System
- ML – Machine Learning
- REST – Representational State Transfer
- SQL – Structured Query Language
- UML – Unified Modeling Language
- UI – User Interface
- UX – User Experience
- XSS-Cross-Site-Scripting

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CHAPTER ONE: INTRODUCTION

1.1 Background of the Project

University libraries play a crucial role in supporting teaching, learning, and research by providing access to books, journals, and digital resources. However, many university libraries, particularly in developing countries, still rely on manual or semi-automated systems to manage their operations.

These traditional systems are often inefficient and result in several challenges, including slow and time-consuming book search processes, difficulties in tracking borrowed and returned materials, increased workload for librarians, and limited user interaction. Consequently, users experience delays in accessing resources, which reduces overall service quality.

With the advancement of Artificial Intelligence (AI) and modern software technologies, there is an opportunity to transform traditional libraries into intelligent and efficient systems. AI techniques enable systems to analyze user behavior, such as borrowing history and search patterns, and provide personalized services such as book recommendations.

Therefore, this project proposes an AI-Based Smart Library Management and Recommendation System to automate library operations, improve resource accessibility, and enhance overall user experience.

1.2 Statement of the Problem

Despite the importance of libraries in supporting academic activities, the current library system at Mekdela Amba University faces several operational challenges. The existing processes are largely manual or semi-digital, which makes them inefficient and time-consuming.

In particular, book searching is slow and inefficient, which delays access to academic resources. In addition, the system lacks a personalized recommendation mechanism that can support users in discovering relevant books based on their interests and

borrowing history. Tracking borrowed and returned books is also difficult, which may lead to inconsistencies in records and reduced data accuracy.

Furthermore, librarians experience a high administrative workload due to repetitive manual tasks such as record updating and user management. Although existing systems support basic functions such as cataloging and borrowing, they do not incorporate intelligent features that can enhance user experience and decision-making.

Therefore, there is a clear need for an automated and intelligent library system that improves operational efficiency and integrates AI-based recommendation capabilities to enhance resource accessibility and user satisfaction.

1.3 Objectives of the Project

1.3.1 General Objective

The main objective of this project is to develop an AI-Based Smart Library Management and Recommendation System that automates library operations and provides personalized book recommendations to enhance user experience and resource accessibility.

1.3.2 Specific Objectives

- To automate core library operations, including book cataloging, borrowing, and returning processes in order to reduce manual workload and improve efficiency.
- To design and implement an efficient and fast book search system that enables users to easily locate academic resources.
- To develop an AI-based recommendation engine that analyzes user behavior and provides personalized book suggestions.
- To design a user-friendly interface that enhances usability and ensures smooth interaction between users and the system.
- To generate reports and analytics for librarians to support decision-making and system monitoring.
- To enhance system security and data management through proper authentication and controlled access mechanisms.

1.4 Scope and Limitation of the Project

1.4.1 Scope of the Project

The system focuses on:

- ❖ Library management at Mekdela Amba University
- ❖ Book management operations including adding, updating, and deleting records
- ❖ User registration and authentication
- ❖ Borrowing and returning system
- ❖ AI-based recommendation system for personalized book suggestions
- ❖ Web-based access for both students and library staff

1.4.2 Limitation of the Project

- Recommendation accuracy depends on the availability and quality of user data
- No integration with external or inter-library systems
- Requires a stable internet connection to operate effectively
- Limited to academic resources only

1.5 Significance of the Project

The AI-Powered Library Management System is significant because it improves the efficiency and effectiveness of library operations by automating essential tasks such as book cataloging, circulation, and user management. By reducing manual administrative work, the system enables librarians to focus on providing better services to users. The user-friendly interface allows students and staff to quickly search for and access library resources, enhancing the overall user experience. Furthermore, the integration of artificial intelligence provides personalized book recommendations based on users' borrowing history and interests, helping them discover relevant learning materials more easily. The system also supports academic success and research activities by ensuring timely access to appropriate resources, making the library more responsive to the needs of its users.

1.6 Target Beneficiaries

The proposed system is designed to benefit the following groups:

- Students: Gain faster access to library resources and receive personalized book recommendations that support learning and academic success.
- Lecturers and Researchers: Efficiently discover relevant books, journals, and academic materials for teaching, research, and knowledge development.
- Librarians: Reduce manual workload, improve operational efficiency, and simplify the management of library resources and services.
- University Administration: Support digital transformation initiatives by modernizing library services, improving information management, and enhancing resource utilization.

1.7 Feasibility Analysis

Feasibility analysis evaluates whether the proposed AI-Based Smart Library Management and Recommendation System is practical, achievable, and suitable for implementation at Mekdela Amba University. The analysis considers operational, technical, economic, behavioral, and schedule feasibility to determine the viability of the proposed system.

1.7.1 Operational Feasibility

The proposed system is operationally feasible for Mekdela Amba University because it is designed to support and enhance the university's library services and resource management activities. The system enables students, academic staff, librarians, and administrators to efficiently search for books, borrow and return materials, access digital resources, and receive personalized book recommendations.

The system is expected to reduce manual record-keeping, minimize processing time, improve service delivery, and enhance the overall user experience. Furthermore, since users are already familiar with digital platforms and information systems used within educational environments, the adoption of the proposed system is expected to require minimal training and support.

1.7.2 Technical Feasibility

The proposed AI-Based Smart Library Management and Recommendation System is technically feasible because Mekdela Amba University has access to the necessary computing infrastructure, internet connectivity, and software resources required to support a web-based application. The system is proposed to be developed using React.js for frontend development, Django (Python) for backend services, and MySQL for database management.

To support intelligent book recommendations, the system will utilize machine learning and data processing libraries such as NumPy, Pandas, and Scikit-learn. The recommendation engine will employ a content-based filtering approach with cosine similarity to generate personalized book suggestions based on users' borrowing history and interests.

Communication between the frontend, backend, database, and recommendation module will be achieved through REST APIs, ensuring efficient data exchange and system integration. Since all proposed technologies are widely used, open-source, and compatible with the university environment, the development and deployment of the system are considered technically achievable.

1.7.3 Economic Feasibility

The proposed AI-Based Smart Library Management and Recommendation System is economically feasible because it utilizes open-source technologies, which significantly reduce software acquisition and licensing costs. The major technologies required for system development, including React.js, Django, Python libraries, and MySQL, are freely available and widely supported by the development community.

In addition, Mekdela Amba University already possesses the essential infrastructure, such as computers, internet connectivity, and server facilities, required to support the implementation and operation of the system. As a result, the project requires minimal additional investment and can be implemented at a relatively low cost.

Therefore, the proposed system is considered cost-effective and financially viable for the university.

1.7.4 Behavioral / Political Feasibility

The proposed AI-Based Smart Library Management and Recommendation System is behaviorally and politically feasible because it supports Mekdela Amba University's commitment to digital transformation and improved academic resource management. The system is designed to simplify library operations, enhance access to information resources, and provide intelligent services that benefit students, academic staff, and librarians.

Users are expected to readily accept the system due to its user-friendly design, improved efficiency, and enhanced accessibility to library services. Furthermore, the proposed system aligns with the university's strategic goals of modernizing educational services, promoting technology-supported learning, and improving institutional efficiency. Therefore, strong organizational support and user acceptance are anticipated during implementation.

1.7.5 Schedule Feasibility

The proposed AI-Based Smart Library Management and Recommendation System is scheduled feasible because the project can be completed within the planned development timeframe. The development process is organized into well-defined phases, including requirements analysis, system design, implementation, testing, and deployment.

By following this structured development approach, project activities can be effectively planned, monitored, and completed according to the project schedule. The availability of development tools, resources, and project documentation further supports the timely completion of the proposed system. Therefore, the project is considered achievable within the allocated timeframe.

1.8 Methodology for the Project

This section describes the methods, approaches, and tools used in the analysis and design of the proposed AI-Based Smart Library Management and Recommendation System. The methodology includes requirement gathering, system analysis and design, technology selection, database design, and system modeling activities.

1.8.1 Requirement Gathering Methods

To accurately identify and define the system requirements, the following requirement gathering techniques were employed:

- Interviews: Structured discussions were conducted with librarians, students, and academic staff to identify existing challenges, user needs, and expected system functionalities.
- Questionnaires: Questionnaires were distributed to library users and staff to collect feedback regarding current library operations, service quality, and desired system improvements.
- Observation: The existing library workflow and operational processes were observed to understand real-world activities and identify inefficiencies in the current system.

These techniques helped ensure the collection of accurate, complete, and reliable requirements, which formed the foundation for the analysis and design of the proposed system.

1.8.2 System Analysis and Design Methodology

The proposed AI-Based Smart Library Management and Recommendation System is analyzed and designed using standard software engineering methodologies and modeling techniques. These approaches help ensure a structured, organized, and maintainable system design.

The following techniques and tools are used during the analysis and design process:

- Unified Modeling Language (UML) Diagrams: Used to model the system's structure, functionality, and interactions. The diagrams include Use Case, Class, Activity, Sequence, Component, and Deployment Diagrams.
- Entity Relationship (ER) Diagram: Used to design the database structure and define relationships among system entities such as users, books, borrowing records, and recommendations.

- ❖ Agile Methodology: Adopted to support iterative and incremental development, allowing continuous refinement and adaptation to changing requirements throughout the project lifecycle.
- ❖ Diagrams.net (Draw.io): Used to create and document UML and ER diagrams for system analysis and design. Available at: https://app.diagrams.net/

This methodology promotes flexibility, continuous improvement, effective system modeling, and better alignment with user and organizational requirements.

1.8.3 Development Tools and Technology Stack

The proposed AI-Based Smart Library Management and Recommendation System utilizes modern web development technologies and machine learning tools to ensure scalability, reliability, and efficient performance.

The following technologies are proposed for system development:

- ❖ Frontend: React.js is used to develop an interactive, responsive, and user-friendly interface that enhances user experience across different devices.
- ❖ Backend: Django Framework is used to implement business logic, manage system operations, and provide RESTful APIs for communication between system components.
- ❖ Database: MySQL is used for storing and managing structured data, including user information, book records, borrowing transactions, and recommendation data.
- ❖ Artificial Intelligence: Python-based libraries such as NumPy, Pandas, and Scikit-learn are used to implement the recommendation engine and support data processing and analysis activities.

These technologies provide a robust foundation for developing a secure, scalable, efficient, and maintainable library management system capable of supporting intelligent recommendation services.

1.9 Risk, Assumptions, and Constraints

This section outlines the key risks, assumptions, and constraints considered during the development of the system. These factors are important in understanding the limitations and operational context of the proposed system.

Risks

The following risks are identified in the development of the system:

- Limited availability and quality of data for training the AI-based recommendation system, which may affect recommendation accuracy.
- Possible system performance issues under heavy usage or high concurrent user access.
- Potential integration challenges between the frontend, backend, and AI modules during system development.

Assumptions

The development of the system is based on the following assumptions:

- Users have access to internet connectivity and digital devices such as computers or smartphones.
- Library data is available in a structured format or can be digitized for system use.
- Users are familiar with basic digital system operations or can adapt with minimal training.

Constraints

The system development is subject to the following constraints:

- Limited development time available for project completion.
- Limited technical and hardware resources for advanced model training and deployment.
- Dependence on open-source tools and technologies due to budget limitations.

1.10 Team Composition and Roles

The project is developed by a team with clearly defined roles and responsibilities to ensure effective collaboration and successful system implementation.

- Meseret Belay: Responsible for documentation, frontend development and system implementation, with a focus on designing and developing the user interface.
- Eliyas Kindie: Responsible for backend development, including server-side logic implementation, API development, and supporting documentation activities.
- Hana Yaregal: Responsible for documentation, frontend design and interface development, ensuring usability, responsiveness, and a user-friendly system interaction experience. Also works on project documentation.

Each team member is assigned specific modules of the system, and their collaboration ensures proper integration of frontend, backend, and supporting components for the successful completion of the project.

1.11 Project Schedule and Milestone

The project is developed through a series of structured phases to ensure systematic progress and timely completion. Each phase contributes to the overall software development lifecycle of the system.

The key milestones of the project include:

Requirement Analysis: Identification and gathering of system requirements through interviews, questionnaires, and observation to determine the functional and non-functional needs of the system.

System Design: Design of the system architecture, database structure, and user interface using UML diagrams and ER diagrams.

Implementation: Development of the system using React.js for the frontend, Django for the backend, MySQL for database management, and AI-based components for the recommendation system.

Testing: Execution of unit testing, integration testing, and system testing to ensure functionality, reliability, and performance of the system.

Documentation: Preparation of complete project documentation covering analysis, design, implementation, testing, and results.

Each phase is executed in a sequential manner to ensure smooth development progress, minimize risks, and achieve timely completion of the project.

1.12 Organization of the Project

The project documentation is organized into six main chapters, each addressing a specific phase of the system development lifecycle. Each chapter is structured to provide a clear and logical flow of the project work from introduction to conclusion.

Chapter One: Introduction – Presents the background of the project, problem statement, objectives, scope, significance, feasibility analysis, and overall methodology of the system.

Chapter Two: Literature Review and Existing System Analysis – Reviews related works, analyzes the current library system, identifies limitations, and justifies the need for the proposed system.

Chapter Three: System Analysis and Modeling – Describes system requirements and models the system using UML diagrams, including use case, class, and behavioral diagrams.

Chapter Four: System Design – Presents the detailed system architecture, database design, user interface design, security design, and system deployment structure.

Chapter Five: Implementation and Testing – Explains system development using selected technologies and presents testing strategies, test cases, and evaluation results.

Chapter Six: Conclusions and Recommendations – Summarizes the project outcomes, highlights challenges faced, and provides recommendations for future improvement.

CHAPTER TWO: LITERATURE REVIEW AND EXISTING SYSTEM ANALYSIS

2.1 Introduction

This chapter presents a review of related literature and an analysis of existing library systems. It examines previous studies and technologies used in library management systems, identifies their limitations, and highlights areas that require improvement.

Furthermore, the chapter evaluates the current library system in use, discusses its major shortcomings, and provides justification for the development of the proposed AI-Based Smart Library Management and Recommendation System.

2.2 Review of Related Works / Literature Review

Library management systems have evolved over time to improve the efficiency and effectiveness of library operations. Initially, libraries relied on traditional manual systems that used paper-based records for managing books and users. These systems were time-consuming, error-prone, and inefficient in handling large volumes of data.

To overcome these limitations, digital library management systems such as Koha and Evergreen were introduced. These systems automated core library functions, including cataloging, borrowing, returning, and user management, thereby significantly improving operational efficiency and reducing manual workload.

However, despite these improvements, most existing systems primarily focus on basic resource management and lack intelligent capabilities. They are unable to analyze user behavior such as borrowing patterns, search history, and preferences, and therefore cannot provide personalized recommendations.

Recent advancements in Artificial Intelligence (AI) and Machine Learning (ML) have introduced recommendation systems that enhance user experience by providing personalized suggestions. Common approaches include:

- ❖ Collaborative Filtering: Recommends books based on the preferences and behavior of similar users, such as borrowing history and ratings.
- ❖ Content-Based Filtering: Recommends books based on item features such as title, author, category, and keywords.
- ❖ Hybrid Recommendation System: Combines collaborative and content-based filtering to improve recommendation accuracy and overcome limitations of individual methods.

These techniques are widely applied in domains such as e-commerce platforms, online learning systems, and streaming services to improve user satisfaction and engagement.

Despite these advancements, many existing library management systems have not fully integrated AI-based recommendation techniques. This creates a significant gap in current solutions.

Therefore, the proposed system aims to address this gap by integrating an AI-based recommendation engine into the library management system, enabling personalized book suggestions and improving overall user experience.

2.2.1 Review of Related Research Works

Several researchers have contributed to the development of intelligent library management and recommendation systems using Artificial Intelligence and Machine Learning techniques.

Smith et al. (2021) proposed a collaborative filtering-based recommendation system for digital libraries. Their system analyzes user borrowing history to recommend similar books based on user behavior patterns. However, the system suffers from the cold-start problem, where new users receive low-quality recommendations due to insufficient data.

Johnson and Lee (2020) developed a content-based book recommendation system that uses metadata such as title, author, and category to suggest similar books. While the system performs well for individual preferences, it lacks the ability to consider behavior from other users, reducing recommendation diversity.

Kumar et al. (2022) introduced a hybrid recommendation model combining collaborative filtering and content-based filtering to improve accuracy in academic library systems. Their results showed better performance compared to single-method systems; however, the model required high computational resources and large datasets.

Ahmed et al. (2023) designed an AI-powered smart library system integrating machine learning techniques for personalized recommendations. The system improved user satisfaction by analyzing search history and borrowing patterns, but it faced limitations in real-time scalability.

Zhang and Wang (2024) proposed an intelligent digital library system using deep learning-based recommendation techniques. The system achieved high accuracy in predicting user interests but required complex training and high computational power.

2.3 Overview of the Existing System

Currently, many university libraries, including Mekdela Amba University, operate using either manual or semi-digital systems. Manual systems rely on paper-based records and register books to manage library activities, while semi-digital systems use basic databases that provide limited automation without intelligent capabilities.

These systems support essential library operations such as book registration, user registration, borrowing and returning of books, and basic search functionality. They ensure that core services are available to users; however, they remain limited in efficiency and functionality.

However, these systems lack advanced features such as intelligent data processing, user behavior analysis, and personalized services. As a result, although they are functional, they are not efficient in meeting modern digital library requirements and user expectations.

2.4 Functionality of the Existing System

The existing library system provides basic functionalities that support day-to-day library operations. These functions are categorized into different operational areas as follows:

- Book Management: The system allows adding new books, updating existing book information, and removing books from the library database.
- User Management: The system supports user registration and maintains basic user information for library access.
- Borrowing System: The system handles issuing books to users and records borrowing and return dates.
- Basic Search: Users can search for books using simple criteria such as title or author.

Although these functionalities support essential library operations, they are limited in efficiency and lack advanced capabilities such as automation, data analytics, and personalized recommendation features. As a result, the system does not fully meet modern library management requirements.

2.5 Business Rules of the Existing System

The existing library system operates based on a set of predefined rules that govern user behavior and system operations to ensure proper library management and control. These rules include:

- Users must be registered in the system before being allowed to borrow books.
- Each user is permitted to borrow only a limited number of books at a time, based on library policy.
- Borrowed books must be returned within a specified time period defined by the library.
- Late returns may result in penalties or restrictions according to library regulations.
- Only books marked as available in the system can be borrowed by users.

These rules provide basic control over library operations and ensure proper resource management. However, they are limited in flexibility and do not include intelligent decision-making or automated enforcement mechanisms.

2.6 Drawbacks, Bottlenecks, and Shortcomings

Despite supporting basic library operations, the existing system has several limitations that affect its efficiency, usability, and overall performance. These limitations are categorized as follows:

2.6.1 Performance (Response Time)

The system experiences slow response times during key operations. Book searching is inefficient, and both borrowing and returning processes take additional time due to limited automation and lack of system optimization.

2.6.2 Input and Output Problems

Manual data entry increases the likelihood of human errors, resulting in inaccurate or outdated information. Additionally, the search functionality is limited and inflexible, making it difficult for users to retrieve relevant information efficiently.

2.6.3 Security and Control Issues

The system has weak authentication mechanisms, which reduces overall security. Manual and semi-digital data handling increases the risk of data loss, and there is also a possibility of unauthorized access to system information.

2.6.4 Efficiency Issues

The system places a high workload on librarians due to excessive manual processing. Library operations are time-consuming, and the system is not capable of efficiently handling a large number of concurrent users.

2.6.5 Other Problems

The existing system lacks intelligent features such as personalized book recommendations. It also does not provide reporting or analytical tools. As a result, the overall user experience remains limited and is not aligned with modern library management standards.

2.7 Justification for the Proposed System (Gap Analysis)

Based on the analysis of the existing library system, several gaps have been identified between the current system and modern library management requirements. These limitations highlight the need for a more intelligent, efficient, and user-centered solution.

The comparison between the existing system and the proposed system is presented below:

Existing System Limitations	Proposed System Improvements
Relies heavily on manual operations	Automates major library processes, including book management, borrowing, and returning activities
Slow and inefficient search mechanisms	Provides fast and advanced search functionality for efficient resource discovery
Does not provide personalized recommendations	Implements an AI-based recommendation engine to suggest relevant books based on user interests and borrowing history
Limited user experience and accessibility	Offers a modern, user-friendly, and responsive interface
Lacks reporting and analytical capabilities	Provides reporting and analytics features to support monitoring and decision-making

Table 1: Gap Analysis

The proposed System addresses these limitations by integrating automation, intelligent recommendation services, and enhanced usability. As a result, the system is expected to improve operational efficiency, increase user satisfaction, and support data-driven decision-making within the university library environment.

2.8 Proposed System Advantages

The proposed AI-Based Smart Library Management and Recommendation System offers several advantages over the existing library system:

- Automates library operations such as book management, borrowing, and returning processes, reducing manual effort and processing delays.
- Improves the speed and accuracy of resource discovery through advanced search functionality.
- Provides AI-based personalized book recommendations based on users' interests, borrowing history, and preferences.
- Enhances user experience through a modern, responsive, and user-friendly interface.
- Supports librarians and administrators with reporting and analytics features for monitoring library activities and informed decision-making.

Overall, the proposed system improves operational efficiency, resource accessibility, user satisfaction, and the overall quality of library services.

CHAPTER THREE: SYSTEM ANALYSIS AND MODELING

3.1 Introduction to System Analysis and Modeling

System analysis and modeling involve examining the requirements of a proposed system and representing them using appropriate diagrams and models. This process helps stakeholders understand how the system will operate, interact with users, manage data, and achieve its intended objectives before implementation begins.

In this project, system modeling is used to represent the following key aspects of the proposed system:

- ➢ System functionality and major services provided to users.
- ➢ User interactions and relationships with different system components.
- ➢ Data flow and information processing within the system.
- ➢ Overall system structure, behavior, and operational workflow.

These models serve as a blueprint for the proposed AI-Based Smart Library Management and Recommendation System, ensuring a clear understanding of system requirements, structure, and functionality before development and implementation.

3.2 Overview of the Proposed System

The proposed AI-Based Smart Library Management and Recommendation System is designed to improve library operations and enhance user experience through the integration of intelligent technologies. The system combines traditional library management functions with Artificial Intelligence (AI) features to provide efficient, personalized, and user-centered services.

The key intelligent components of the proposed system include:

- ❖ AI Recommendation Engine: Analyzes users' borrowing history, search activities, and reading preferences to generate personalized book recommendations using content-based filtering techniques.

- ❖ Library Management Module: Supports core library operations such as book management, user management, borrowing, returning, and digital resource access.
- ❖ Reporting and Analytics Module: Generates reports and analytical insights to support monitoring, resource management, and decision-making activities.

These components work together to improve accessibility, automation, operational efficiency, and overall user satisfaction within the university library environment.

3.3 Requirements of the Proposed System

3.3.1 Functional Requirements

Functional requirements describe the services and operations that the proposed system must perform to meet user and organizational needs. The proposed AI-Based Smart Library Management and Recommendation System shall:

- ➢ Support user registration, authentication, and login management.
- ➢ Allow librarians to add, update, delete, and manage book records.
- ➢ Enable users to search for books by title, author, category, or keyword.
- ➢ Manage book borrowing and returning transactions efficiently.
- ➢ Track borrowed books and monitor their availability and status.
- ➢ Generate reports on library activities, book circulation, and user transactions.
- ➢ Provide AI-based personalized book recommendations based on users' interests and borrowing history.

These functional requirements define the core services and capabilities of the proposed system and serve as a foundation for system design and development.

3.3.2 Non-Functional Requirements

Non-functional requirements define the quality attributes, performance standards, and operational constraints of the proposed system. These requirements ensure that the system functions efficiently, securely, and reliably.

Performance

The system should provide fast response times for user authentication, book searches, borrowing transactions, and recommendation generation. It should also support multiple users simultaneously without significant performance degradation.

Security

The system must ensure secure authentication and authorization mechanisms. User information, library records, and transaction data should be protected from unauthorized access through appropriate security measures and secure communication protocols.

Usability

The system should provide a simple, intuitive, and user-friendly interface that enables users to perform tasks efficiently with minimal training and technical knowledge.

Reliability

The system should operate consistently and accurately while ensuring data integrity, availability, and correctness. It should minimize system failures and maintain stable operation under normal usage conditions.

Scalability

The system should support future growth in the number of users, books, digital resources, and system features without significantly affecting performance or functionality.

Maintainability

The system should be designed in a modular and organized manner to facilitate maintenance, error correction, system updates, and future enhancements.

3.4 Use Case Modeling

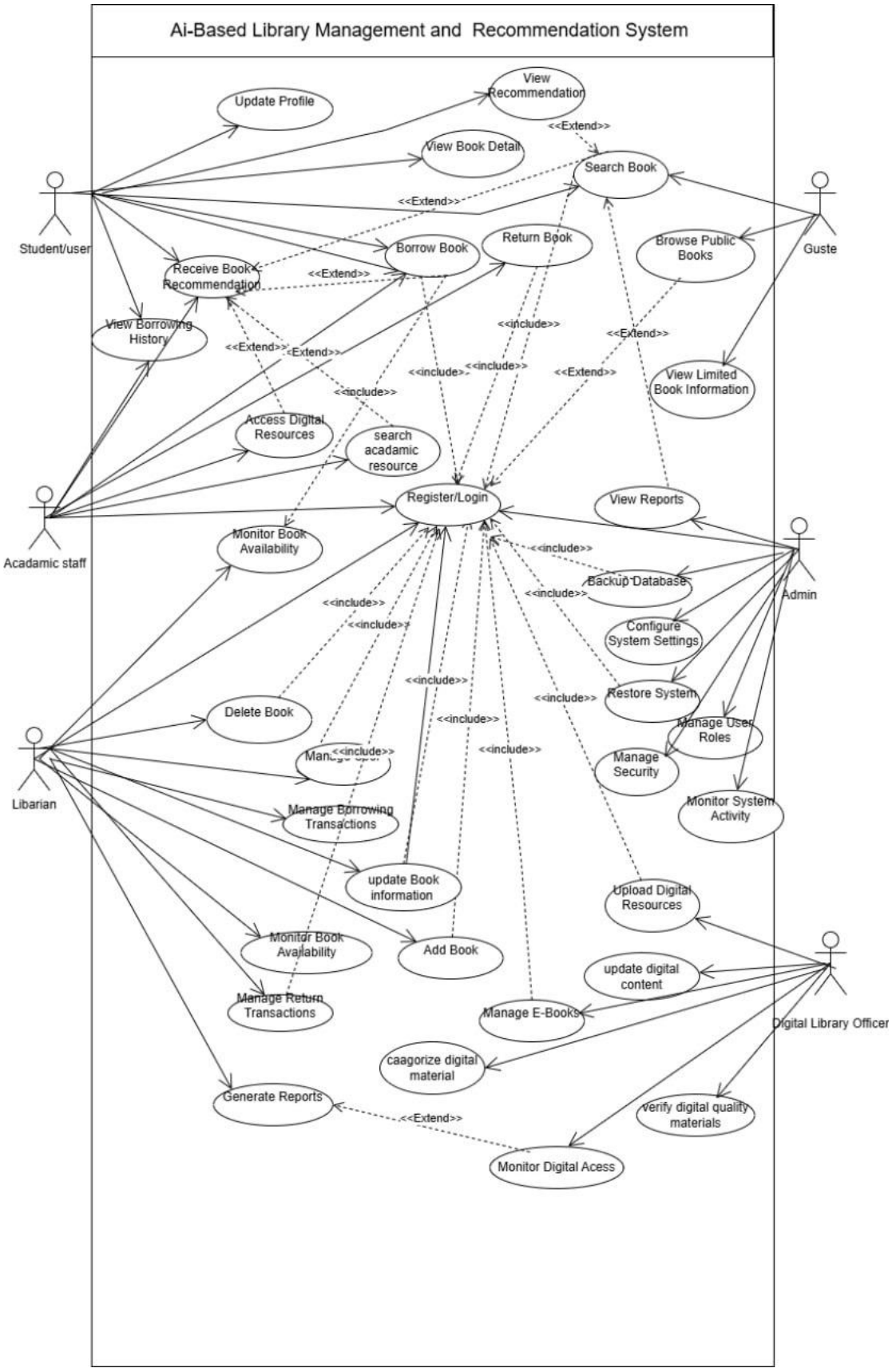


Figure 1:Usecase Diagram

3.4.1 Use Case Diagram (Description)

The Use Case Diagram illustrates the interactions between users and the proposed AI-Based Smart Library Management and Recommendation System. It identifies the actors involved in the system and the services they can access based on their roles and responsibilities.

The system consists of five primary actors and one optional actor:

- Student / Library User: Searches for books, borrows and returns materials, accesses digital resources, and receives personalized book recommendations.
- Academic Staff: Searches for academic resources, borrows materials, accesses digital content, and receives recommendation services.
- Librarian: Manages books, borrowing and returning transactions, user records, and library operations.
- System Administrator: Manages user accounts, access permissions, system security, and overall system configuration.
- Digital Library Officer: Manages digital resources, uploads electronic content, and ensures the quality and availability of digital materials.
- Guest User: Accesses limited public services such as viewing available books and general library information without authentication.

Each actor interacts with the system according to assigned permissions and responsibilities. The system uses «include» relationships to represent mandatory actions that must occur before another use case can be executed, such as user authentication before borrowing books. It also uses «extend» relationships to represent optional or conditional functionalities, such as generating book recommendations after a search operation or allowing a guest user to register for full system access.

The Use Case Diagram provides a high-level view of system functionality and helps define the interactions between users and the proposed system.

3.5 Analysis-Level Class Diagram (Conceptual Model)

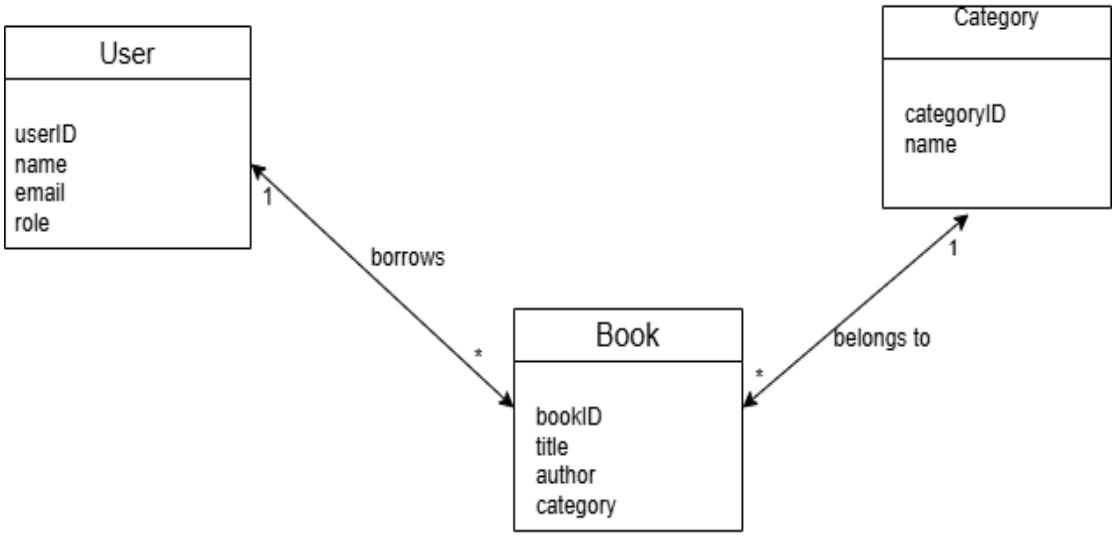


Figure 2:Analaysis Level Diagram

The Class Diagram represents the static structure of the proposed AI-Based Smart Library Management and Recommendation System by illustrating its main classes, attributes, methods, and relationships. It provides an object-oriented view of the system and shows how different entities interact to support library operations and intelligent recommendation services.

In the proposed system, users such as students, academic staff, librarians, and system administrators interact with the system according to their assigned roles and permissions. The Book and Digital Resource classes represent the primary library assets managed by the system. The Borrow class handles borrowing and returning transactions, while the Recommendation class supports AI-based personalization by generating book suggestions based on user interests and borrowing history.

The inclusion of digital resources extends the system beyond traditional physical library services by supporting electronic learning materials and digital content management. Overall, the Class Diagram provides a comprehensive representation of the system structure and demonstrates how traditional library functionalities are integrated with modern intelligent services to enhance efficiency and user experience.

3.6 Key Abstraction with CRC(Class-Responsibility-Collaboration) Analysis

Key abstraction is the process of identifying the main classes or objects in a system that represent its core components. In object-oriented design, these classes model real-world entities and define the overall system behavior.

CRC (Class–Responsibility–Collaboration) analysis is a technique used to describe:

- ❖ Class: The main object in the system
- ❖ Responsibility: The functions or tasks performed by the class
- ❖ Collaboration: The interaction between classes

This technique helps to clearly define system structure, improve modularity, and ensure proper interaction among system components.

In the proposed AI-Based Smart Library Management and Recommendation System, the main identified classes are: User, Book, Borrow, Recommendation, DigitalResource, and SystemAdmin.

CRC Analysis of Main Classes

1. Class: User (Student, Academic Staff, Librarian, Admin, Digital Library Officer, Guest)

   Responsibilities:

   - ➢ User registration and authentication (login/logout)
   - ➢ Profile management
   - ➢ Role-based access to system functionalities
   - ➢ Viewing personalized recommendations

   Collaborations:

   - Book (searching and viewing books)
   - Borrow (borrowing and returning books)
   - Recommendation (receiving personalized suggestions)
   - DigitalResource (accessing digital content)

- SystemAdmin (account and role management)

2. Class: Book

   Responsibilities:

   - ➢ Manage book information (title, author, category, status)
   - ➢ Maintain availability status (available, borrowed, reserved)
   - ➢ Provide metadata for search and recommendation
   - ➢ Support catalog management

   Collaborations:

   - User (search and access books)
   - Borrow (borrowing and returning transactions)
   - Recommendation (AI-based suggestions)
   - Librarian (book management operations)

3. Class: Borrow (Transaction Class)

   Responsibilities:

   - ➢ Manage borrowing transactions between users and books
   - ➢ Store borrowing details (borrow date, return date, status)
   - ➢ Enforce borrowing rules (limits and due dates)
   - ➢ Maintain borrowing history for analysis

   Collaborations:

   - User (borrower identification)
   - Book (availability updates)
   - Librarian (transaction monitoring)
   - Recommendation (behavior analysis data)

4. Class: Recommendation (AI Module Class)

   Responsibilities:

   - ➢ Analyze user behavior (search history and borrowing patterns)
   - ➢ Generate personalized book recommendations
   - ➢ Apply content-based similarity analysis

- ➢ Rank and update recommendations dynamically

Collaborations:

- User (preference analysis)
- Book (content matching)
- Borrow (historical data analysis)
- DigitalResource (enhanced suggestions)

5. Class: DigitalResource

   Responsibilities:

   - ➢ Manage digital materials (e-books, PDFs, journals)
   - ➢ Organize and store academic digital content
   - ➢ Provide metadata for indexing and search
   - ➢ Support recommendation system enhancement

   Collaborations:

   - Digital Library Officer (upload and management)
   - User (access to digital resources)
   - Recommendation (digital content suggestions)

6. Class: SystemAdmin (Supporting Class)

   Responsibilities:

   - ➢ Manage system users and roles
   - ➢ Monitor system performance and logs
   - ➢ Ensure security and access control
   - ➢ Configure system settings

   Collaborations:

   - User (account and role management)
   - All system modules (monitoring and control)

3.7 Behavioral / Dynamic Model

The behavioral or dynamic model describes how the system behaves over time in response to user actions and system events. It focuses on the interaction between system components and illustrates the sequence of operations required to complete specific tasks.

In the proposed AI-Based Smart Library Management and Recommendation System, behavioral modeling is used to describe how different actors—Student/Library User, Academic Staff, Librarian, System Administrator, and Digital Library Officer—interact with the system. It also shows how internal components such as Book, Borrow, Recommendation, and DigitalResource collaborate during system execution.

This model helps define the flow of control, data exchange, and system responses during key operations such as searching for books, borrowing resources, returning books, and generating AI-based recommendations.

Main Purpose of Behavioral Modeling

The behavioral model is used to:

- ➢ Describe how users interact with the system over time
- ➢ Define system responses to user actions
- ➢ Represent message flow between objects
- ➢ Model real-time processes such as borrowing, returning, and recommending books
- ➢ Ensure proper interaction between frontend, backend, database, and AI components

UML Diagrams Used in Behavioral Modeling

The following UML diagrams are used to represent the dynamic behavior of the system:

- ❖ Sequence Diagram: Shows time-ordered interactions between objects
- ❖ Activity Diagram: Represents the workflow of system processes
- ❖ State Diagram: Describes the lifecycle of objects (e.g., Book state changes)

Key Behavioral Processes in the System

1. User Authentication Process
    - ➢ User enters credentials
    - ➢ System validates identity
    - ➢ Role-based access is granted (Student, Staff, Librarian, Admin)
2. Book Search and View Process
    - ➢ User submits a search request
    - ➢ System retrieves matching records from the database
    - ➢ Results are displayed to the user
    - ➢ Recommendation module may be triggered
3. Borrowing Process (Core Dynamic Flow)
    - ➢ User selects a book
    - ➢ System checks availability
    - ➢ Borrow transaction is created
    - ➢ Book status is updated to “Borrowed”
    - ➢ Confirmation is sent to the user
4. Return Book Process
    - ➢ User returns the borrowed book
    - ➢ System updates borrowing records
    - ➢ Book status is changed to “Available”
    - ➢ User borrowing history is updated
5. AI Recommendation Process
    - ➢ System collects user behavior (search history and borrowing records)
    - ➢ Recommendation engine analyzes data
    - ➢ Similarity scoring is applied
    - ➢ Personalized book recommendations are generated
    - ➢ Results are displayed to the user

The behavioral model ensures that all system components interact in a structured and predictable manner. It clearly defines how the system responds to user actions and how data flows between the user interface, backend services, database, and AI recommendation engine.

This model is essential for ensuring system correctness, smooth user interaction, proper integration of AI components, and efficient execution of library operations.

3.7.1 Sequence Diagram (Example: Borrow Book)

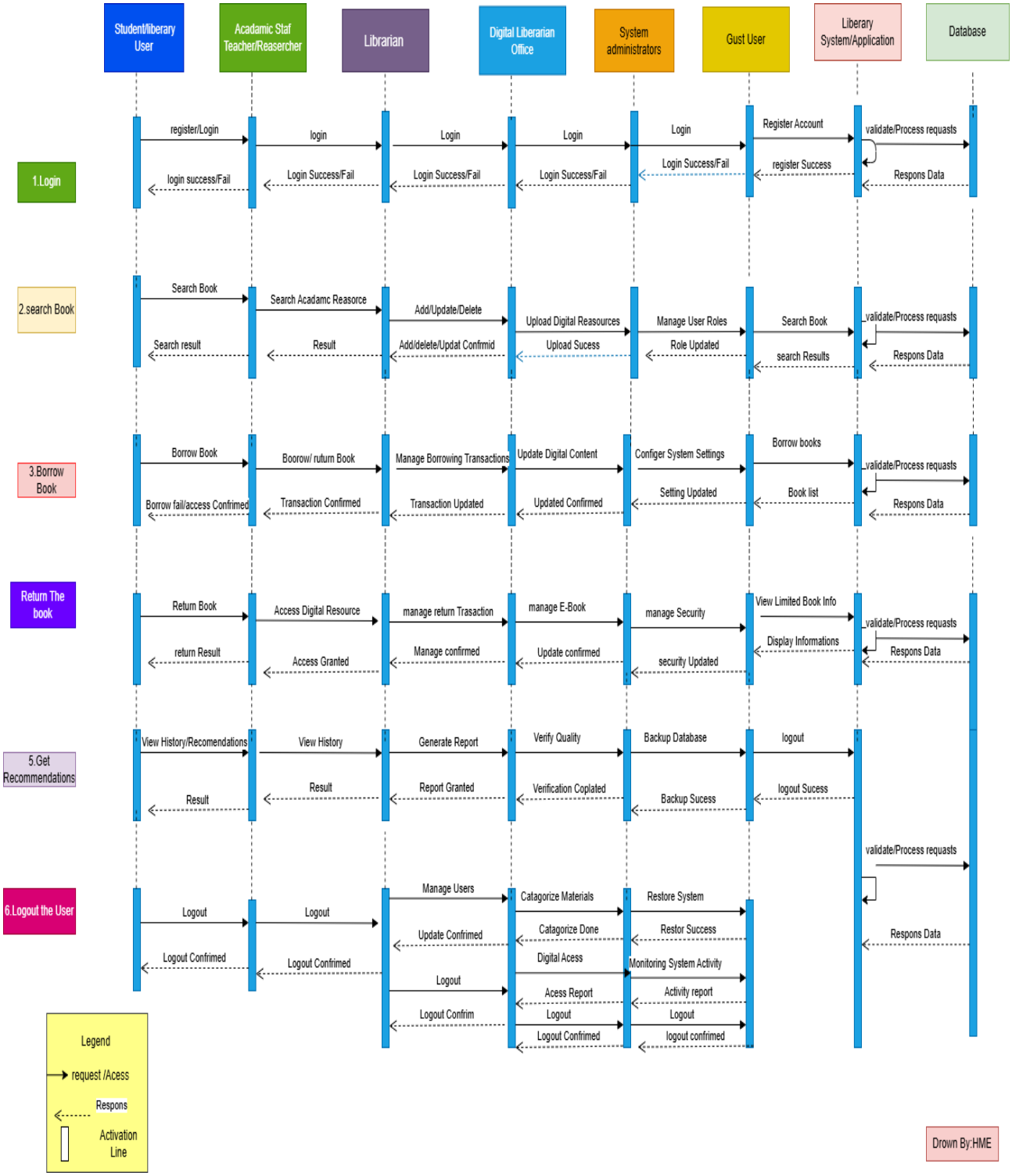


Figure 3:Sequence Diagram

The Sequence Diagram illustrates the interaction and coordination between system components during the book borrowing process. It shows the time-ordered flow of messages exchanged between the user and the system to complete the borrowing operation.

This diagram ensures proper coordination between system components and supports:

- ➢ Data consistency across the system
- ➢ Real-time updates of book availability and transactions
- ➢ Role-based access control for authorized users
- ➢ Integration of AI-based recommendation services
- ➢ Efficient handling of library transactions

Overall, the sequence diagram provides a clear representation of how the borrowing process is executed step by step within the system, ensuring proper communication between the frontend, backend, database, and AI recommendation modules.

3.7.2 Activity Diagram (Flow)

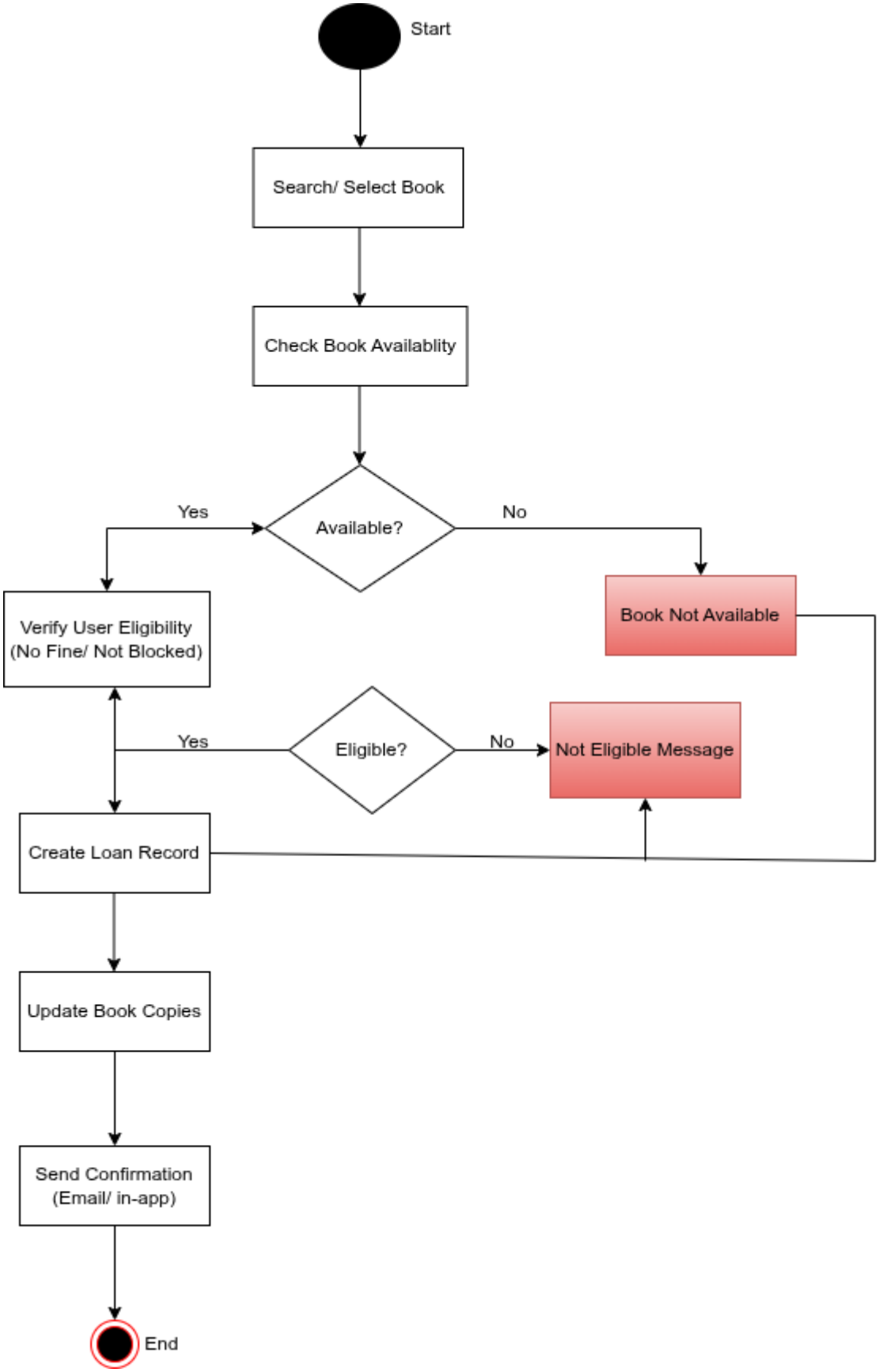


Figure 4:Activity Diagram

The Activity Diagram illustrates the workflow of the system and shows how the

system responds to different conditions during the borrowing process. It provides a clear visualization of the sequence of activities performed from the initial request to the completion of the transaction.

This diagram helps to improve:

- ➢ Process clarity and workflow understanding
- ➢ Error handling and decision control
- ➢ User experience through structured flow design
- ➢ System automation and process optimization
- ➢ Integration of AI-based recommendation features during execution

Overall, the activity diagram ensures that the system processes are well-organized, logically structured, and efficiently executed, enabling smooth interaction between users and system components.

3.7.3 State Diagram

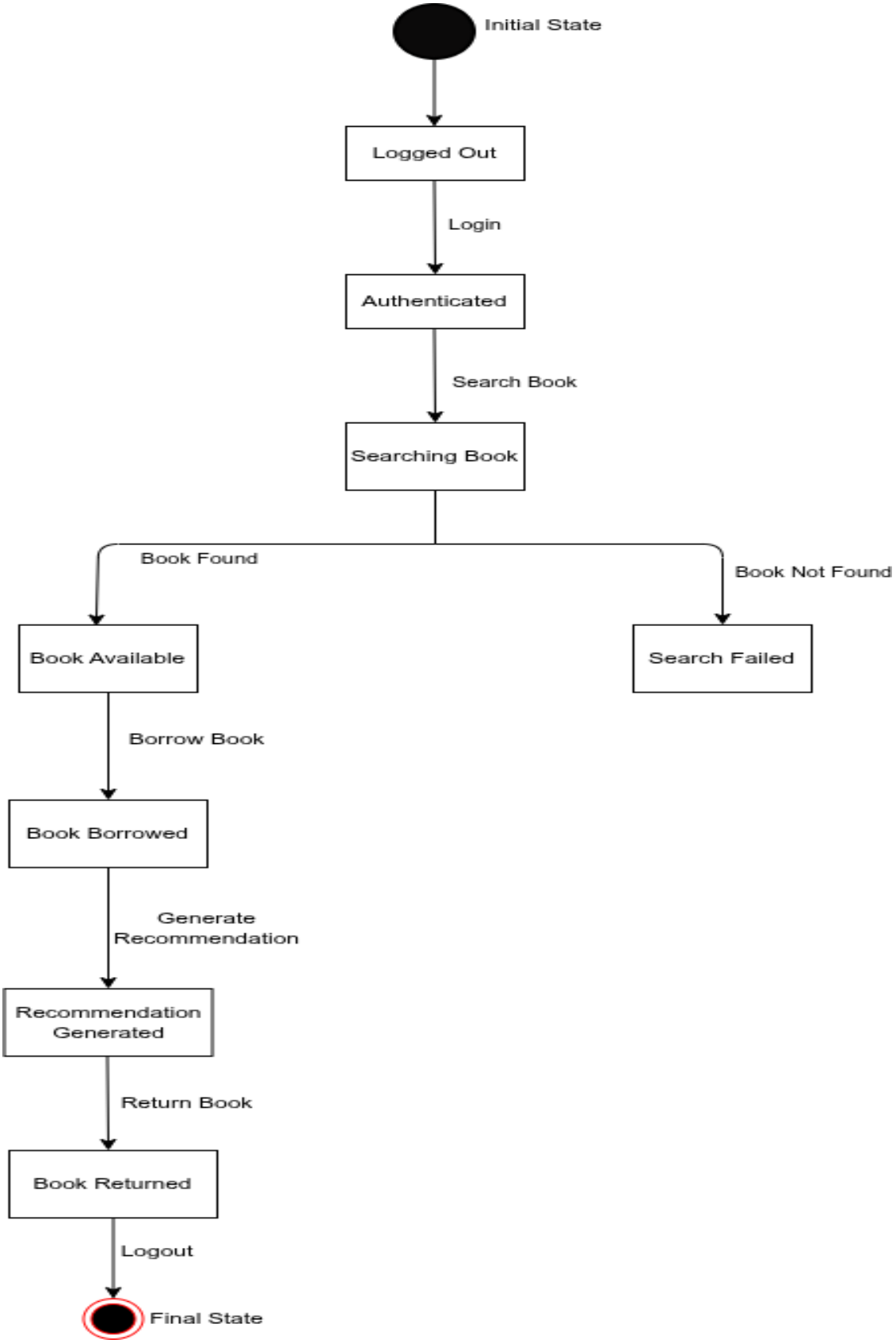


Figure 5:State Diagram

The State Diagram describes the different states of objects within the AI-Based Smart Library Management and Recommendation System and how these states change in response to system events.

In this system, the lifecycle of a book and borrowing process is modeled based on state transitions triggered by user actions. The process begins when a user logs into the system and searches for a book. The system then checks the database to determine whether the requested book exists.

If the book is not found, the system remains in the search state and prompts the user to refine their query. If the book is found, the system transitions to a verification state where the availability of the book is checked. When the book is unavailable, the system moves to an “unavailable” state and notifies the user accordingly. If the book is available, the system transitions to a “borrow processing” state, where borrowing permissions are validated and the transaction is recorded.

After successful borrowing, the system updates the book state to “borrowed” and triggers the AI recommendation module, which generates personalized suggestions. Finally, the system returns to the idle or ready state after displaying a confirmation message.

This state-based modeling ensures proper tracking of object lifecycles, consistent system behavior, and efficient management of library operations.

3.8 Supplementary Specifications

In addition to the core functional and non-functional requirements, the system includes several supplementary features that enhance usability, reliability, and overall system performance. These features ensure that the system operates efficiently across all user roles, including Student/Library Users, Academic Staff (Teachers/Researchers), Librarians, System Administrators, Digital Library Officers, and Guest Users.

Additional System Features:

1. Error Handling

   The system provides comprehensive error handling mechanisms to ensure smooth operation and user-friendly interaction. It addresses issues such as:

   - ➢ Invalid login attempts across all user roles (Students, Staff, Librarians, and Administrators)
   - ➢ Unauthorized access attempts (e.g., Guest Users attempting restricted operations such as borrowing)
   - ➢ Selection of unavailable books during borrowing processes
   - ➢ System and database connection errors

   Clear and descriptive error messages are displayed to guide users in correcting their actions and improving system usability.

2. Notification System

   The system includes an automated notification module designed to improve communication and user awareness. Notifications are sent to:

   - ➢ Students and Academic Staff for due date reminders and overdue alerts
   - ➢ Librarians for monitoring borrowing activities and book returns
   - ➢ System Administrators for system events and security alerts

   These notifications help reduce late returns and enhance overall library efficiency.

3. Recommendation Accuracy Improvement

   The AI-based recommendation engine is continuously improved by analyzing user behavior across different roles such as Students and Academic Staff. It utilizes borrowing history, search patterns, and category preferences to:

   - ➢ Improve personalization of book recommendations
   - ➢ Increase relevance of suggested materials
   - ➢ Adapt recommendations to different academic disciplines

The Digital Library Officer supports data quality management, which contributes to improving recommendation accuracy over time.

3.9 AI Recommendation Model

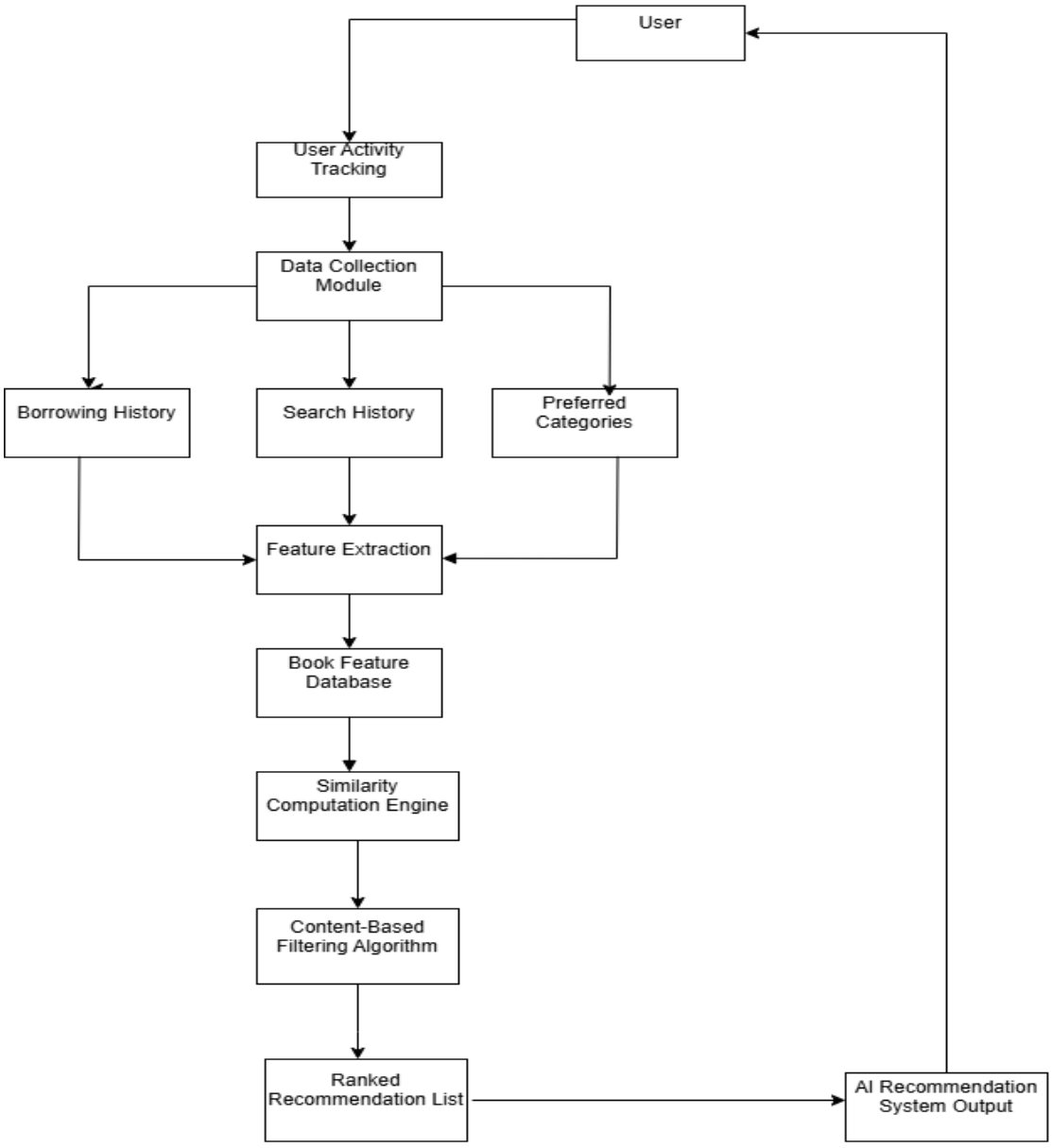


Figure 6:AI Recommendation Diagram

The system collects user data such as borrowing history, search activities, and preferences from Student/Library Users and Academic Staff (Teachers/Researchers). This data is securely managed under the supervision of the System Administrator,

while the Digital Library Officer ensures that book metadata and resources are accurate, consistent, and well-organized.

The recommendation engine processes this data using multiple approaches, including Collaborative Filtering, Content-Based Filtering, and Hybrid Recommendation Systems. These techniques analyze user behavior and item characteristics to identify relationships between users and books.

- Collaborative Filtering is used to recommend books based on the behavior and preferences of similar users.
- Content-Based Filtering recommends books by analyzing similarities between book attributes and user interests.
- Hybrid Recommendation System combines both approaches to improve accuracy and overcome limitations of individual methods.

Based on these techniques, the system generates a ranked list of relevant book recommendations, which is displayed to users.

Additionally, the Librarian can utilize these recommendations to support decision-making in library resource management and collection development.

CHAPTER FOUR: SYSTEM DESIGN

4.1 Introduction to System Design

System design is the phase in the software development life cycle where requirements identified during system analysis are transformed into a structured blueprint for implementation. It defines how the system will be built by specifying its architecture, components, data structures, database design, and user interface.

In this project, the design supports multiple system actors, including Student/Library Users, Academic Staff (Teachers/Researchers), Librarians, System Administrators, Digital Library Officers, and Guest Users (optional). Each actor has different access levels and interactions, which are reflected in the system design.

The primary objective of system design is to provide a clear and well-structured framework that guides developers during implementation. It ensures that the system is efficient, secure, scalable, and maintainable while supporting role-based access control and AI-based recommendation functionality.

By translating abstract requirements into concrete design models, system design serves as a critical foundation for building a reliable and high-quality smart library system. It defines the overall structure of the system, including how users interact with the system and how different modules such as book management, borrowing, authentication, and recommendation work together.

4.2 Overall System Architecture

The proposed system follows a three-tier architecture, which divides the system into three main layers. This separation improves scalability, security, and maintainability. The architecture also supports multiple system actors, including Student/Library Users, Academic Staff (Teachers/Researchers), Librarians, System Administrators, Digital Library Officers, and Guest Users, with role-based access control.

1. Presentation Layer (Frontend)

   This layer represents the user interface of the system and is developed using React.js. It enables users to interact with the system through web interfaces such as login, registration, dashboards, book search, borrowing pages, and recommendation interfaces. Each user role accesses system features according to assigned permissions.

2. Application Layer (Backend)

   This layer handles the business logic of the system and is implemented using the Django framework. It processes user requests, enforces system rules, and manages core functionalities such as authentication, book management, borrowing transactions, and recommendation generation. It also ensures secure communication between the frontend and the database while enforcing role-based access control.

3. Data Layer (Database)

   This layer is responsible for storing and managing all system data using MySQL. It maintains structured data such as user information, book records, borrowing transactions, and AI-generated recommendations. The System Administrator and Digital Library Officer are responsible for ensuring data integrity, security, and proper data management.

Key Advantages of the Architecture

- ➢ Improved scalability to support a large number of users and resources
- ➢ Enhanced security through controlled access and authentication mechanisms
- ➢ Easier maintenance and modular system development
- ➢ Efficient communication between system components
- ➢ Support for role-based access control and AI-based recommendation features

4.3 Subsystem Decomposition

The system is divided into several smaller subsystems to simplify development, improve maintainability, and support different system actors, including Student/Library Users, Academic Staff (Teachers/Researchers), Librarians, System Administrators, Digital Library Officers, and Guest Users.

1. User Management System

   This subsystem handles all user-related operations, including registration, login, authentication, and role management. It supports multiple user roles with different access levels, which are controlled and managed by the System Administrator.

2. Book Management System

   This subsystem manages all book-related operations such as adding, updating, deleting, and viewing books. It is primarily managed by the Librarian, while the Digital Library Officer supports data organization, categorization, and quality assurance.

3. Borrowing and Circulation Management System

   This subsystem handles borrowing, returning, and renewal operations. It tracks borrowing transactions, maintains borrowing history, enforces borrowing limits and due dates, and supports overdue book monitoring and fine management.

4. Recommendation System (AI Module)

   This subsystem generates personalized book recommendations based on user behavior, including search history, borrowing patterns, and preferences. It supports Students and Academic Staff by suggesting relevant academic materials using Artificial Intelligence techniques.

5. Reporting and Analytics System

This subsystem generates reports and analytical summaries related to library usage, borrowing trends, popular resources, overdue items, and inventory status. It supports monitoring, planning, and decision-making activities.

Together, these subsystems provide an integrated framework that supports both traditional library management functions and intelligent recommendation services.

4.4 Layered / Class-Type Architecture

4.4.1 User Interface Layer

This layer includes all web pages such as login, registration, dashboard, book search, borrowing, and recommendation pages. It is developed using React.js and is responsible for displaying information to users and collecting input from different system actors, including Students, Academic Staff, Librarians, and Guest Users (optional).

4.4.2 Controller / Process Layer

This layer processes all requests coming from the user interface. It acts as an intermediary between the frontend and backend services. It handles request routing, input validation, and communication with business logic components. It also ensures that user actions are properly authorized based on roles defined by the System Administrator.

4.4.3 Business Layer

This layer contains the core business logic of the proposed AI-Based Smart Library Management and Recommendation System. It manages user authentication, borrowing and returning operations, recommendation generation, and library management rules. The layer also enforces borrowing policies, access control mechanisms, and recommendation processing to ensure that system operations are performed consistently and securely..

4.4.4 Persistence Layer

This layer handles all database operations such as creating, reading, updating, and deleting records. It manages data related to users, books, borrowing transactions, and recommendations using a MySQL database. It ensures data consistency, integrity, and secure storage under the supervision of the System Administrator.

4.4.5 System Layer

This layer includes the underlying infrastructure that supports the entire system operation. It consists of the operating system, web server, database server, network infrastructure, and hardware resources required to run the application.

It ensures reliable communication between the frontend, backend, and database layers, and supports deployment of the system across different environments. This layer is maintained and monitored by the System Administrator, ensuring system availability, performance, and security for all actors, including Students, Academic Staff, Librarians, and Digital Library Officers.

4.5 Design Class Diagram

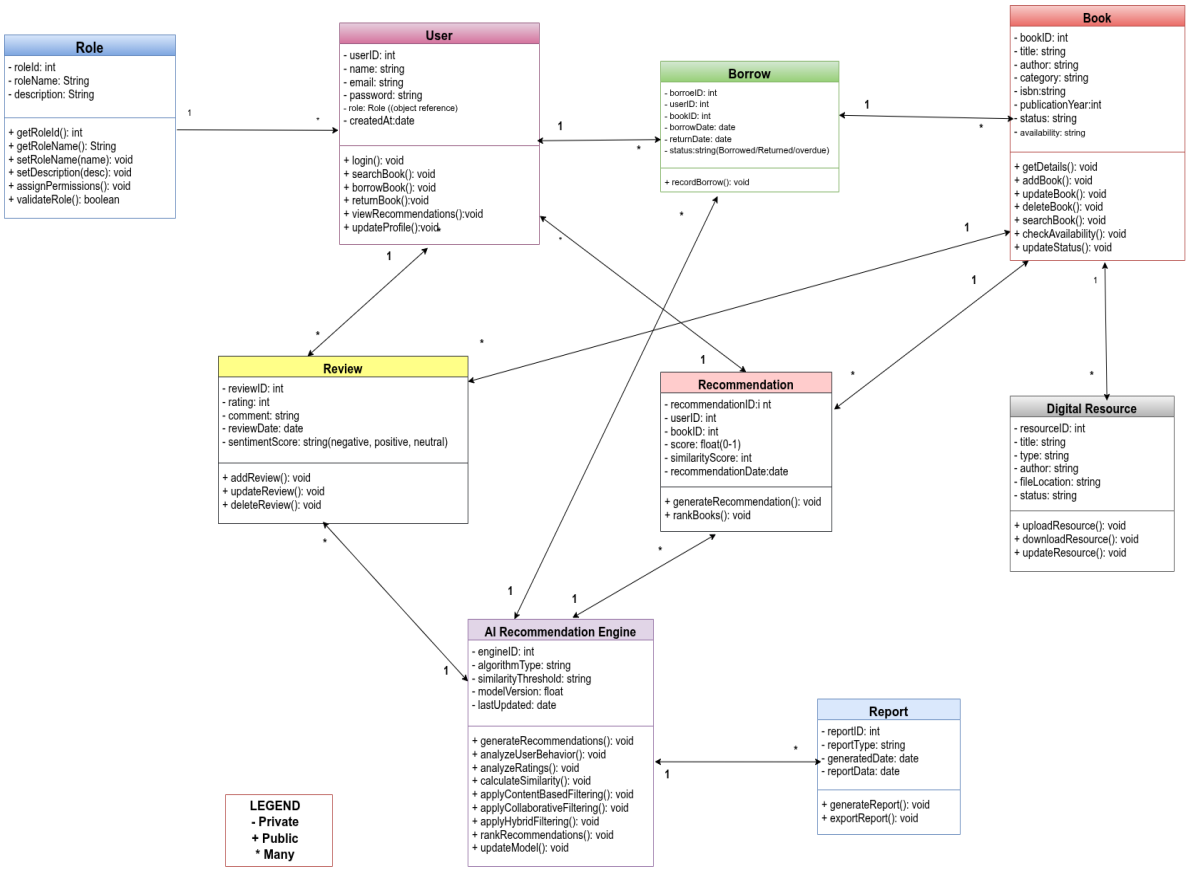


Figure 7:Design Class Diagram

The Design Class Diagram illustrates the structural organization of the AI-Based Smart Library Management and Recommendation System. It shows the main classes, their attributes, operations, and the relationships between them, providing a clear blueprint for system implementation.

The system consists of several key classes: User, Book, Borrow, Review, AI Recommendation Engine, Recommendation, DigitalResource, and Report. The User class interacts with the system by managing authentication, profile information, searching books, borrowing resources, and receiving recommendations. The Book class represents library resources and manages book details, availability, and search operations.

The Borrow class links the User and Book classes and records borrowing transactions such as borrow date, due date, return date, and status. The Review class allows users to provide ratings and comments on books, which helps improve recommendation accuracy. The AI Recommendation Engine class analyzes user behavior, borrowing

history, and reviews to generate personalized recommendations, which are stored in the Recommendation class.

The DigitalResource class manages electronic materials such as e-books and journals, while the Report class collects data from all major modules to generate analytical reports for decision-making.

Overall, the diagram demonstrates how users, library resources, borrowing activities, reviews, digital content, and AI-based recommendation components interact to form an intelligent and efficient library management system.

4.6 Collaboration Modeling

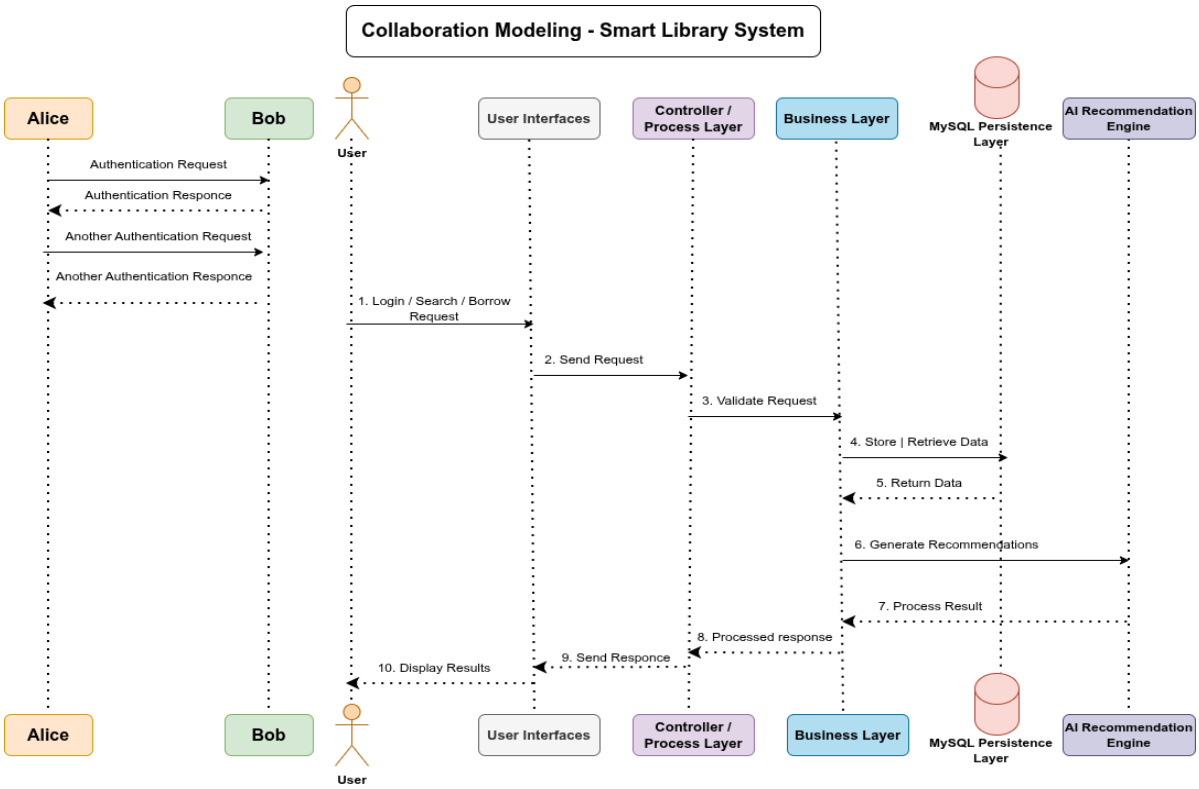


Figure 8:Collaborative Diagram

The Collaboration Model illustrates how system components interact to support the operations of the AI-Based Smart Library Management and Recommendation System. Users, including students, academic staff, librarians, administrators, digital library officers, and guest users, interact with the system through the User Interface Layer.

User requests are forwarded to the Controller/Process Layer for validation and routing. The Business Layer then processes core functionalities such as authentication, role-based access control, borrowing management, and AI-based recommendation generation. The Persistence Layer (MySQL Database) stores and retrieves system data, including user profiles, books, borrowing records, reviews, and recommendations.

After processing, the results are returned through the system layers to the User Interface Layer and presented to the user. This collaboration ensures secure, efficient, and coordinated communication among all system components.

4.7 Component Diagram

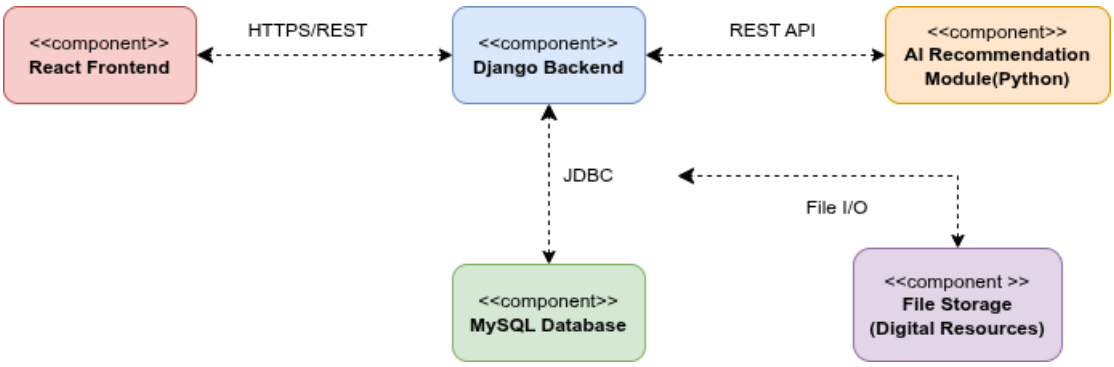


Figure 9:Component Diagram

The Component Diagram illustrates the major software components of the AI-Based Smart Library Management and Recommendation System and their interactions. The system consists of five main components: the React Frontend, Django Backend API, MySQL Database, AI Recommendation Engine, and Digital Resource Storage.

The React Frontend provides the user interface for accessing system services, while the Django Backend API manages business logic, authentication, role-based access control, and request processing. The MySQL Database stores and manages user accounts, book records, borrowing transactions, reviews, and recommendation data. The AI Recommendation Engine analyzes user behavior and borrowing history to generate personalized book recommendations. The Digital Resource Storage component manages electronic resources such as e-books, journals, research papers, and other digital documents.

These components communicate through APIs to ensure secure, efficient, scalable, and intelligent library management services.

4.8 Deployment Diagram

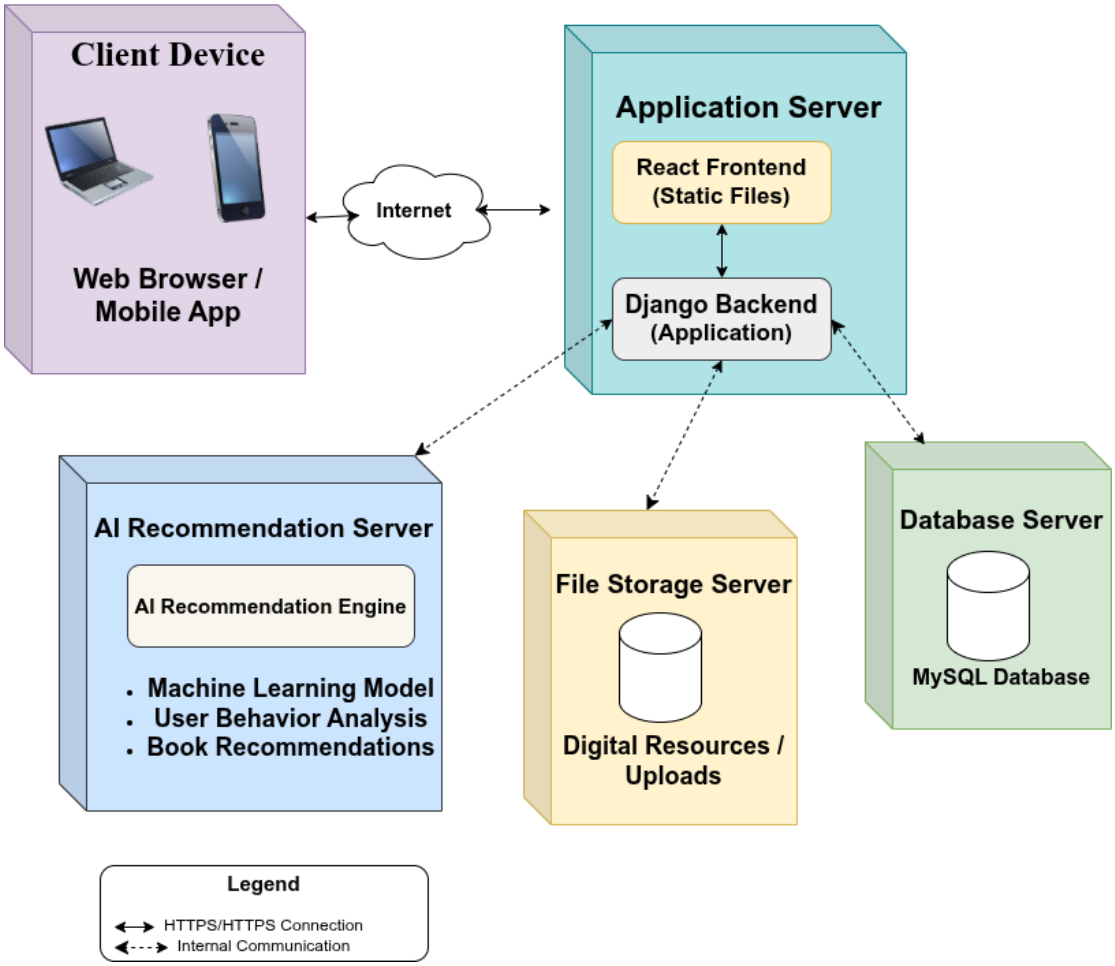


Figure 10:Deployment Diagram

The Deployment Diagram illustrates the physical architecture of the AI-Based Smart Library Management and Recommendation System and shows how system components are deployed across hardware nodes.

The system consists of a Client Device, Application Server, Database Server, AI Recommendation Server, and Digital Resource Storage Server. Users access the system through a web browser or mobile device on the Client Device. Requests are sent to the Application Server, which hosts the frontend and backend services and manages system operations.

The Application Server communicates with the Database Server to store and retrieve user information, book records, borrowing transactions, reviews, reports, and recommendation data. It also interacts with the AI Recommendation Server to

generate personalized book recommendations and with the Digital Resource Storage Server to access and manage digital resources such as e-books, journals, research papers, and other electronic documents.

This deployment architecture provides secure, scalable, reliable, and efficient system operation while supporting intelligent library management and recommendation services.

4.9 Database Design / Persistence Modeling

4.9.1 Entity Relationship Diagram (ERD)

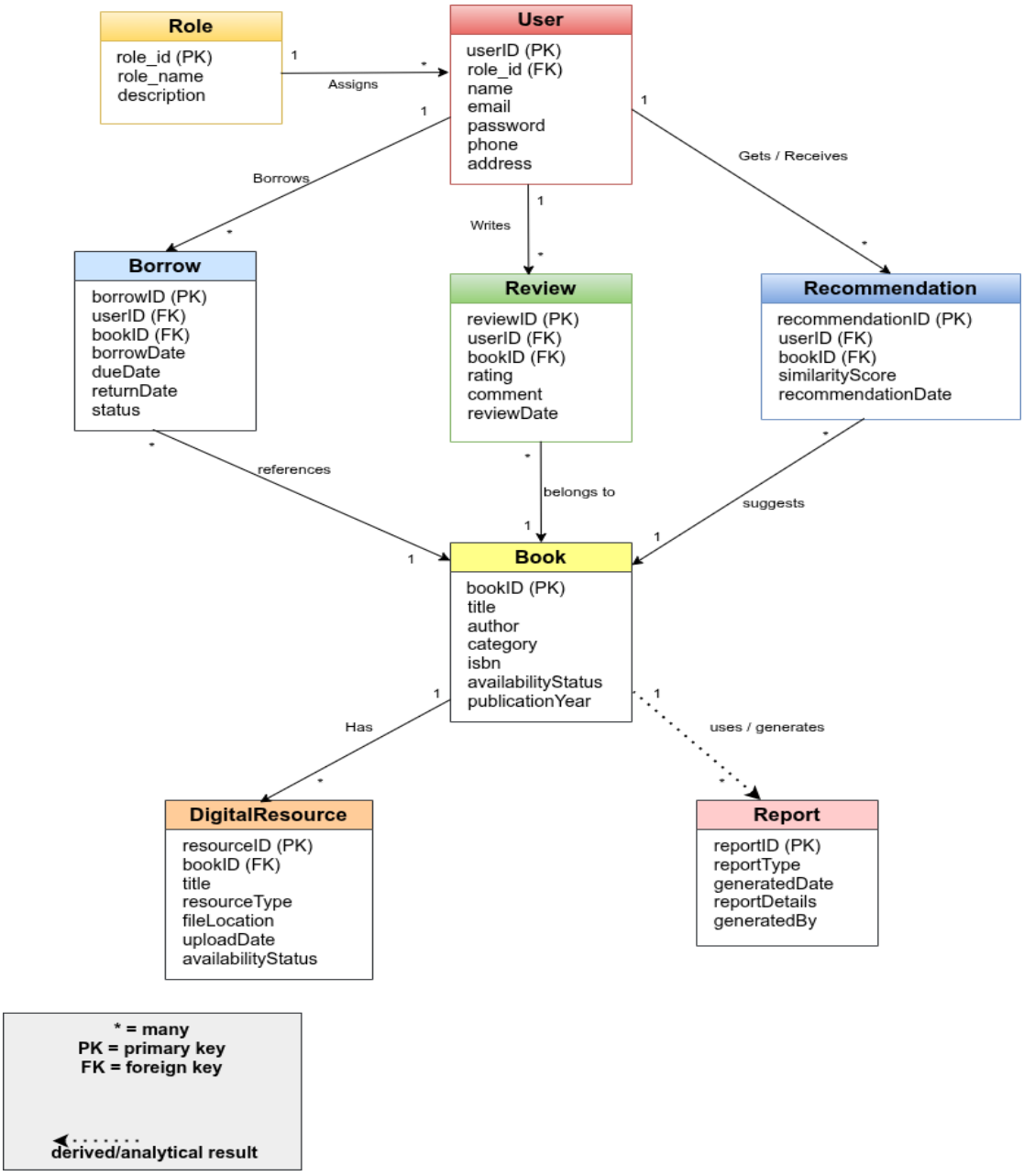


Figure 11:ER Diagram

The ER diagram illustrates the structural design of the Library Management System by defining its main entities and their relationships. It models how users interact with library resources through borrowing, reviewing, and AI-based recommendations, ensuring efficient data organization and consistency.

The system consists of the following core entities:

- ❖ User: Stores user details such as userID, name, email, password, and role. It represents all system users with different access levels.
- ❖ Book: Stores book information including bookID, title, author, category, and availability status. It represents the main library resources.
- ❖ Borrow: Captures borrowing transactions between users and books, including borrow date, due date, return date, and status.
- ❖ Recommendation: Stores AI-generated book suggestions for users, including similarity scores based on user behavior and preferences.
- ❖ DigitalResource: Represents electronic materials such as e-books and journals linked to books.
- ❖ Report: Provides analytical summaries generated from system data for decision-making.

Relationships

A User can borrow many Books, and each Book can be borrowed by many Users through the Borrow entity. Users can also write reviews and receive AI-based recommendations for books. Each Book can have multiple DigitalResources. Reports are generated from overall system data including users, books, borrowing activities, reviews, and recommendations.

4.9.2 Database Schema (Tables)

The database schema defines the logical structure of data storage in the Library Management System using relational tables. Each table represents a core system entity and supports all actors, including Student/Library Users, Academic Staff, Librarian, System Administrator, Digital Library Officer, and Guest Users. Primary keys ensure unique identification, while foreign keys maintain relationships between tables and ensure data integrity.

User Table

The User table stores information about all system users.

- ➢ user_id (PK): Unique identifier for each user
- ➢ name: Full name of the user

- email: Unique email address used for login
- password: Encrypted password for authentication
- role: Defines user type (Student, Academic Staff, Librarian, Admin, Guest)

Book Table

The Book table stores details of all library books.

- book_id (PK): Unique identifier for each book
- title: Title of the book
- author: Author of the book
- category: Subject or genre
- status: Availability status (Available, Borrowed, Unavailable)
- added_by: References the librarian or system admin who added the book

Borrow Table

The Borrow table records book borrowing transactions and links users with books.

- borrow_id (PK): Unique transaction ID
- user_id (FK): References User table
- book_id (FK): References Book table
- borrow_date: Date borrowed
- return_date: Date returned
- due_date: Return deadline
- status: Borrow status (Borrowed, Returned, Overdue)

Recommendation Table

The Recommendation table stores AI-generated book suggestions.

- rec_id (PK): Unique recommendation ID
- user_id (FK): References User table
- book_id (FK): References Book table
- score: Similarity score for ranking recommendations
- generated_at: Timestamp of recommendation creation

Role Table

The Role table defines user access levels in the system and supports role-based access control (RBAC).

- role_id (PK): Unique identifier for each role
- role_name: Name of the role (Student, Academic Staff, Librarian, Admin, Guest)
- description: Description of permissions and responsibilities for each role

4.10 User Interface Design (Wireframes)

The user interface design defines the visual structure and layout of the system through wireframes, ensuring intuitive and efficient interaction between users and the system. It supports all system actors, including Student/Library Users, Academic Staff, Librarian, System Administrator, Digital Library Officer, and Guest Users, through role-based access to system features. The interface is designed to ensure usability, accessibility, and responsiveness across desktops, tablets, and mobile devices.

Main Interfaces

Login Page

Allows registered users to securely access the system using their credentials. Authentication is managed by the system, and all actors except Guest Users must login.

Registration Page

Enables new users (Students and Academic Staff) to create accounts by providing personal information. User accounts may be verified by the System Administrator or Librarian.

Dashboard

Acts as the main control center after login, providing access to features such as book search, borrowing, returns, and recommendations. The layout varies based on user role.

Book Search Page

Allows users to search and filter books by title, author, category, or keywords. Guest Users may access this feature with limited permissions.

Recommendation Page

Displays AI-generated personalized book suggestions. It is mainly available for Students and Academic Staff, while Librarians and Admins can view analytics.

Design Principles

- ➢ Simplicity: Clean and minimal interface for easy navigation
- ➢ Responsiveness: Works across desktop, tablet, and mobile devices
- ➢ User-Friendliness: Clear buttons, labels, and feedback for smooth interaction

4.11 Security Design and Access Control

The system incorporates multiple security mechanisms to protect user data, prevent unauthorized access, and ensure system integrity. These mechanisms are designed based on the CIA security model (Confidentiality, Integrity, and Availability).

The key security features include:

- ❖ User Authentication

  The system provides a secure login mechanism that verifies user credentials before granting access. Passwords are stored in encrypted (hashed) form to protect sensitive information.

- ❖ Role-Based Access Control (RBAC)

  System access is restricted based on user roles such as Student/Library User, Academic Staff, Librarian, System Administrator, Digital Library Officer, and Guest User. Each role is assigned specific permissions to ensure controlled access to system features.

- ❖ Data Encryption

Sensitive data, including passwords and user information, is protected using encryption techniques during storage and transmission to prevent unauthorized access.

- Input Validation and Sanitization

  All user inputs are validated and sanitized to prevent security threats such as SQL injection, cross-site scripting (XSS), and invalid data entry.

- Session Management

  Secure session handling is implemented, and inactive users are automatically logged out to reduce the risk of unauthorized access.

- Access Logging and Monitoring

  System activities such as login attempts, transactions, and administrative actions are recorded to support auditing and detect suspicious behavior.

The combined implementation of these security mechanisms ensures a secure, reliable, and controlled environment for all system users.

4.12 Global Control Flow

The global control flow describes how the AI-Based Smart Library Management and Recommendation System processes user requests and manages system operations. The system combines procedural and event-driven approaches to ensure efficient execution of library services and responsive user interaction. After successful authentication, users are granted access to system features according to their assigned roles and permissions.

4.12.1 Procedural-Driven Control Flow

The procedural control flow follows a structured and sequential execution of system operations. After authentication, users perform activities in a predefined order, ensuring consistency and proper management of library transactions.

Typical Procedural Flow:

Login → Role-Based Dashboard → Search Book → View Book Details → Borrow/Return Book → Generate Recommendations → Logout

This approach helps maintain data integrity, enforce borrowing policies, and ensure that system activities are processed in a controlled and predictable manner.

4.12.2 Event-Driven Control Flow

The system also supports an event-driven control flow, where operations are triggered by user actions such as button clicks, form submissions, menu selections, and search requests.

For example, clicking the Search button initiates a book query request, selecting a book triggers availability checking, and requesting recommendations activates the AI recommendation module. Each event generates the appropriate backend process through REST API communication.

This event-driven approach enables real-time responses, improves system usability, and provides a dynamic user experience. By combining procedural and event-driven control flows, the system achieves both structured execution of core processes and flexible interaction with users.

4.13 Boundary Conditions and Exception Handling

This section defines the operational limits of the AI-Based Smart Library Management and Recommendation System and the mechanisms used to handle errors and unexpected situations. Proper management of boundary conditions and exceptions ensures system reliability, data integrity, security, and a smooth user experience.

4.13.1 Boundary Conditions

Boundary conditions define the limits within which the system operates correctly and efficiently.

- Maximum Borrow Limit

The system restricts the number of books a user can borrow simultaneously according to library policies. This ensures fair resource utilization and prevents misuse.

- Book Availability Constraint

  Only books marked as Available can be borrowed. If a book is already borrowed or unavailable, the system blocks the transaction and notifies the user.

- User Access Restriction

  Users can only access functionalities permitted by their assigned roles, such as Student, Academic Staff, Librarian, System Administrator, Digital Library Officer, or Guest User.

- Session Validity Limit

  User sessions automatically expire after a specified period of inactivity to enhance security and prevent unauthorized access.

4.13.2 Exception Handling

The system is designed to detect, manage, and recover from errors while providing clear feedback to users.

- Invalid Login Credentials

  If a user enters an incorrect username or password, the system displays an error message and denies access until valid credentials are provided.

- Book Not Available

  If a user attempts to borrow an unavailable book, the system notifies the user and prevents the transaction.

- Input Validation Errors

The system validates all user inputs and displays appropriate error messages for empty fields, invalid formats, or incomplete information.

- System and Database Errors

  Unexpected system failures or database errors are handled through error logging and controlled responses to maintain system stability and prevent data loss.

These boundary conditions and exception-handling mechanisms improve system reliability, maintain data consistency, and ensure uninterrupted operation under both normal and exceptional conditions.

CHAPTER FIVE: IMPLEMENTATION AND TESTING

5.1 Implementation

The implementation phase focuses on developing the AI-Based Smart Library Management and Recommendation System based on the design specifications presented in Chapter Four. The system is implemented as a web-based application using modern software development technologies and industry best practices.

During this phase, all major modules are developed and integrated, including user management, book management, borrowing and returning services, digital resource management, and the AI-based recommendation module. The frontend, backend, and database components communicate through RESTful APIs to ensure efficient data exchange and system interoperability.

An iterative development approach is followed, where each module is implemented, tested, and refined incrementally to improve functionality, reliability, and performance.

5.1.1 Development Tools and Technologies

The following tools and technologies were used in the development of the system:

- Frontend
  - React.js – Used to build a responsive, interactive, and user-friendly interface.
- Backend
  - Django Framework – Used to implement server-side logic, business rules, and RESTful APIs.
- Database
  - MySQL – Used for secure and structured storage of system data.
- Programming Languages
  - JavaScript – Used for frontend development.
  - Python – Used for backend development and AI recommendation processing.
- AI Libraries and Tools

- ➢ Scikit-learn – Used to implement recommendation algorithms.
- ➢ NumPy – Used for numerical computations and data manipulation.
- ➢ Pandas – Used for data processing and analysis.

❖ Development Tools

- ➢ Visual Studio Code – Source code editor.
- ➢ Git – Version control and collaboration tool.
- ➢ Postman – API testing and debugging tool.

These technologies provide a reliable, scalable, and efficient environment for developing, testing, and maintaining the AI-Based Smart Library Management and Recommendation System.

5.1.2 System Modules Implementation

The system is implemented using a modular approach to improve scalability, maintainability, and separation of concerns. Each module is developed independently and later integrated into the overall system.

User Management Module

The User Management Module handles user registration, login, and authentication processes. It securely stores user information and manages role-based access control for different users, including students, academic staff, librarians, system administrators, and digital library officers.

Book Management Module

The Book Management Module is responsible for adding, updating, deleting, and managing book records. It also maintains the availability status of books to ensure accurate tracking and efficient management of library resources.

Borrowing Module

The Borrowing Module manages book borrowing and returning activities. It records borrowing transactions, maintains borrowing history, enforces borrowing limits, and updates book availability status in real time.

Recommendation Module (AI)

The Recommendation Module utilizes artificial intelligence and machine learning techniques to analyze user behavior, including borrowing history and search patterns. Based on this analysis, the system generates personalized book recommendations that help users discover relevant learning materials.

Reporting Module

The Reporting Module generates various reports such as most borrowed books, active users, overdue books, and library usage statistics. These reports assist librarians and administrators in monitoring system performance and making informed decisions.

5.1.3 Implementation Approach

The AI-Powered Library Management System was developed using the Agile software development approach. This approach supports flexibility, continuous improvement, and iterative development throughout the project lifecycle. It enables the development team to implement features incrementally, test them regularly, and incorporate feedback to improve system quality and performance.

Incremental Development

The system was developed in small and manageable modules. Each module, such as user management, book management, borrowing, recommendation, and reporting, was implemented and tested independently before being integrated into the overall system. This approach helped reduce complexity and ensured that each component functioned correctly.

Iterative Enhancement

System features were continuously refined and improved based on testing outcomes and identified requirements. Through multiple development iterations, enhancements were made to improve functionality, performance, and user experience.

Continuous Integration

Newly developed components were integrated regularly into the main system. This ensured seamless communication and interaction among the frontend, backend, database, and AI recommendation modules while minimizing integration issues.

Feedback-Driven Improvement

Feedback obtained from users, supervisors, and testing activities was used to identify weaknesses and areas for improvement. The collected feedback guided modifications and enhancements, resulting in a more reliable, efficient, and user-friendly system.

This approach ensures flexibility, faster development, and better alignment of the system with user requirements.

5.2 Testing

Testing ensures that the system functions correctly, meets user requirements, and performs reliably under different conditions. It is carried out at different levels to detect and fix errors early before deployment.

5.2.1 Types of Testing

1. Unit Testing

    Tests individual components or modules of the system independently.

    Example: Testing the login function to verify correct authentication behavior.

2.Integration Testing

    Tests the interaction between different modules to ensure they work together correctly.

    Example: Verifying communication between the login module and the database.

3.System Testing:

    Tests the complete system as a whole to ensure all features function correctly in an integrated environment. Example: Checking user login, book search, borrowing, and recommendations together.

4. User Acceptance Testing (UAT):

Conducted by real users such as students and librarians to confirm that the system meets user needs and expectations before final deployment. Example: Students and librarians test the system to verify that book searching, borrowing, returning, and recommendation features work as expected.

Test Case	Input	Expected Output	Result
Login	Correct username/password	Login success	Pass
Login	Wrong password	Error message	Pass
Borrow Book	Available book	Borrow success	Pass
Borrow Book	Unavailable book	Error message	Pass

Table 2:Tasting

5.2.2 Test Cases and Testing Techniques

Test cases are structured scenarios used to verify that the system functions correctly under different conditions. Each test case includes inputs, execution steps, and expected outputs to ensure that the system behaves as intended.

Testing ensures that the system is reliable, secure, and user-friendly. It helps identify errors early, improves system performance, and ensures that the final product meets user requirements.

A. Test Case Structure

A standard test case includes the following components:

- ➢ Test Case ID: Unique identifier for each test case
- ➢ Test Scenario: Description of what is being tested
- ➢ Input Data: Data entered into the system
- ➢ Expected Output: Expected system behavior after execution
- ➢ Actual Output: Result obtained after execution
- ➢ Status: Pass/Fail based on comparison of expected and actual output

B. Functional Test Cases

1. Login Module

Test Case ID	Test Scenario	Input	Expected Output	Status
TC01	Valid Login	Correct username & password	Login successful, dashboard displayed	Pass
TC02	Invalid Password	Correct username, wrong password	Error message displayed	Pass
TC03	Empty Fields	No input	Validation message displayed	Pass
TC04	Invalid Username	Wrong username	Error message displayed	Pass

Table 3: Functional Test Cases

These test cases verify that the login module correctly handles both valid and invalid authentication attempts while ensuring proper validation and user feedback.

2. User Registration

Test Case ID	Test Scenario	Input	Expected Output	Status
TC05	Valid Registration	All correct details	Account created successfully	Pass
TC06	Duplicate Email	Existing email	Error: Email already exists	Pass
TC07	Missing Fields	Empty required fields	Validation error	Pass

Table 4: User Registration

These test cases ensure that the user registration module correctly handles valid inputs, prevents duplicate accounts, and enforces required field validation.

3. Book Search

Test Case ID	Test Scenario	Input	Expected	Status

			Output	
TC08	Search available book	Book title	Book details displayed	Pass
TC09	Search unavailable book	Non-existing title	“Book not found” message	Pass
TC10	Empty search	No input	Prompt to enter search term	Pass

Table 5:Book Searching

These test cases verify that the book search module correctly retrieves available books, handles missing results, and validates empty search inputs.

4. Borrow Book

Test Case ID	Test Scenario	Input	Expected Output	Status
TC11	Borrow available book	Select available book	Borrow successful message	Pass
TC12	Borrow unavailable book	Select unavailable book	Error message displayed	Pass
TC13	Borrow limit exceeded	User reached limit	Warning message displayed	Pass

Table 6:Borrow Book

These test cases ensure that the borrowing module correctly handles successful borrowing, prevents borrowing of unavailable books, and enforces borrowing limits.

5. Return Book

Test Case ID	Test Scenario	Input	Expected Output	Status
TC14	Return borrowed book	Valid return	Return successful	Pass
TC15	Return non-borrowed book	Invalid book	Error message displayed	Pass

Table 7:Return Book

These test cases verify that the return module correctly processes valid returns and prevents invalid return operations.

6. AI Recommendation

Test Case ID	Test Scenario	Input	Expected Output	Status
TC16	Recommend based on history	User with borrowing history	Relevant books recommended based on past activity	pass
TC17	Recommend for new user	New user (no history)	Default or popular books recommended	pass
TC18	Recommend similar category books	User selects a category (e.g., AI)	Books from the same category are recommended	pass
TC19	No matching recommendation	Rare or unique search input	Message: "No recommendations available"	pass
TC20	Multiple preferences	User with mixed interests	Recommendations from multiple related categories	pass
TC21	Recommendation update after borrow	User borrows a new book	Recommendations updated based on new activity	pass

Table 8:AI Recommendation

These test cases verify that the AI recommendation module correctly generates personalized suggestions, handles new and edge-case users, and updates recommendations dynamically based on user activity.

C. Testing Techniques

1. Black Box Testing

   Focuses on testing system functionality without considering internal code structure.

   Example: Verifying login with valid and invalid inputs.

2. White Box Testing

   Tests internal logic and code structure, usually performed by developers.

   Example: Checking conditions and loops in the login function.

3. Boundary Value Analysis

   Tests edge values of input fields to ensure the system handles minimum and maximum limits correctly.

   Example: Minimum and maximum password length validation.

4. Equivalence Partitioning

   Divides input data into valid and invalid groups to reduce the number of test cases while maintaining coverage.

   Example: Valid email formats vs invalid email formats.

5. Error Guessing

   Based on tester experience to predict possible error-prone scenarios in the system.

   Example: Entering special characters or leaving required fields empty.

D. Non-Functional Testing

- ❖ Performance Testing: Ensures the system responds quickly under normal and high load conditions.
- ❖ Security Testing: Verifies protection against unauthorized access and security vulnerabilities.
- ❖ Usability Testing: Ensures the system is easy to learn, navigate, and use for all users.

Testing plays a critical role in ensuring system quality. Through both functional and non-functional testing, the system achieves reliability, security, and efficiency before deployment.

5.2.3 Testing Tools

The following tools and methods are used to test and debug the proposed system effectively:

Manual Testing

Used to verify system functionalities by executing test cases manually. It helps ensure

that features such as login, book searching, borrowing, returning, and recommendations work correctly according to user requirements.

Browser Testing
Used to test the compatibility and responsiveness of the web application across different web browsers such as Google Chrome, Mozilla Firefox, and Microsoft Edge. This ensures consistent system behavior and user experience on different platforms and devices.

Debugging Tools
Debugging tools available in Visual Studio Code, browser developer tools, and Django debugging utilities are used to identify, trace, and fix errors in both frontend and backend components.

API Testing Tools
Tools such as Postman are used to test RESTful APIs developed in Django. These tools verify request handling, response accuracy, authentication, and data communication between frontend and backend modules.

Database Testing Tools
MySQL Workbench is used to test database operations such as data insertion, retrieval, updating, deletion, and relationship validation to ensure database integrity and consistency.

These testing tools help improve system quality, reliability, performance, and overall stability before deployment.

5.2.4 Expected Testing Outcomes

The expected outcomes of the testing process are as follows:

1. Correct System Functionality
   All modules of the system, including login, registration, book management, borrowing, returning, reporting, and AI recommendation features, should operate according to the specified requirements.

2. Reduced Errors and Bugs

   Most system errors, bugs, and unexpected behaviors should be identified and fixed during testing to improve system reliability and stability.

3. Good System Performance

   The system should provide fast response times for searching books, login operations, recommendations, and transaction processing, even when multiple users access the system simultaneously.

4. Secure System Operation

   The testing process should confirm that authentication, authorization, and input validation mechanisms effectively protect the system from unauthorized access and security threats.

5. Accurate AI Recommendations

   The recommendation module should generate relevant and personalized book suggestions based on user preferences, borrowing history, and search activities.

6. User Satisfaction

   Students, academic staff, librarians, and administrators should be able to use the system easily and efficiently through a simple, responsive, and user-friendly interface.

7. Reliable Data Management

   The system should correctly store, retrieve, and manage data without loss, duplication, or inconsistency.

Overall, the testing process ensures that the system is reliable, efficient, secure, and capable of meeting the needs of all users before deployment.

CHAPTER SIX: CONCLUSIONS AND RECOMMENDATIONS

6.1 Conclusions

This project successfully designed and developed an AI-Based Smart Library Management and Recommendation System for Mekdela Amba University. The system addresses the major limitations of the existing manual and semi-digital library system by automating core library operations and integrating intelligent AI-based services.

The system supports multiple actors including Students/Library Users, Academic Staff (Teachers/Researchers), Librarians, System Administrators, Digital Library Officers, and optional Guest Users. Each actor interacts with the system according to assigned roles and permissions, improving overall system management and security.

The major achievements of the project include:

- ➢ Automation of book cataloging, user registration, borrowing, returning, and tracking processes
- ➢ Implementation of a fast and efficient search system with filtering options
- ➢ Development of an AI-based recommendation engine that provides personalized book suggestions based on user interests, borrowing history, and search behavior
- ➢ Design of a responsive and user-friendly web interface using React.js and Django
- ➢ Integration of reporting and analytics features for librarians and administrators
- ➢ Implementation of security mechanisms such as authentication, role-based access control, and input validation
- ➢ Support for digital resource management through the Digital Library Officer role
- ➢ Improved accessibility and interaction for students, researchers, librarians, and administrators

By integrating Artificial Intelligence, database management, and modern web technologies, the proposed system improves operational efficiency, reduces manual workload, enhances user experience, and supports better access to academic resources.

Overall, the project demonstrates how AI and software engineering technologies can modernize university library services and contribute to the digital transformation initiatives of Mekdela Amba University.

6.2 Challenges Faced

During the development of the AI-Based Smart Library Management and Recommendation System, several technical and operational challenges were encountered.

Limited Historical Data for AI Training

The recommendation system required sufficient user borrowing and search history to generate accurate suggestions. Initially, the system had limited data, which affected recommendation accuracy. To reduce this problem, sample datasets and category-based fallback recommendations were used.

Integration Complexity

Integrating the React.js frontend, Django backend, MySQL database, and AI recommendation module was challenging. Proper API communication and data synchronization were required to ensure smooth interaction between all system components.

Time Constraints

As a final-year project, the development time was limited. The team had to prioritize important system features and follow an iterative development approach to complete the project within the scheduled timeframe.

Performance Optimization

Ensuring fast system response during searching, borrowing, and recommendation generation required optimization of database queries and backend processing to support multiple concurrent users efficiently.

Resource Limitations

The project depended mainly on open-source technologies and limited hardware resources. This restricted the use of more advanced AI models and large-scale training processes.

Security and Data Handling Challenges

Implementing secure authentication, role-based access control, and proper data validation required careful design to protect user information and prevent unauthorized access.

User Interface Design Challenges

Designing a responsive and user-friendly interface for different actors such as students, librarians, academic staff, administrators, and digital library officers required additional planning and testing to ensure usability.

Despite these challenges, the project team successfully managed the development process through proper planning, teamwork, continuous testing, and the effective use of open-source technologies. The experience gained improved both technical knowledge and project management skills.

6.3 Recommendations

Although the proposed AI-Based Smart Library Management and Recommendation System successfully fulfills its objectives, several improvements are recommended for future enhancement and expansion.

Mobile Application Development

A cross-platform mobile application using technologies such as React Native can be developed to improve accessibility for students, academic staff, librarians, and administrators through smartphones and tablets.

1. Advanced AI Recommendation Techniques

   The current content-based recommendation system can be upgraded to a hybrid recommendation model by combining content-based and collaborative filtering techniques. Deep learning methods can also be introduced as more user interaction data becomes available.

2. Cloud Deployment and Scalability

   Deploying the system on cloud platforms such as Amazon Web Services, Google Cloud, or Heroku can improve scalability, reliability, backup management, and remote accessibility.

3. Integration with External Digital Libraries

   The system can be integrated with other university libraries, digital repositories, and open educational resource platforms to expand access to academic materials and research resources.

4. Enhanced Notification System

   Future versions of the system should include real-time notifications through email, SMS, or in-app alerts for due dates, overdue books, reservation updates, and newly added resources.

5. User Feedback and Rating Mechanism

   Adding features that allow users to rate books, provide reviews, and submit feedback can improve user engagement and help refine recommendation accuracy.

6. Advanced Security Features

   Additional security mechanisms such as multi-factor authentication (MFA), activity logging, session management, and audit trails can further strengthen system security and data protection.

7. Performance Optimization

   Future improvements should include load testing, database optimization, and caching technologies such as Redis to support a larger number of concurrent users efficiently.

8. AI Chatbot Enhancement

   The AI chatbot assistant can be enhanced with natural language processing capabilities to provide more intelligent responses, voice interaction, and multilingual support for users.

9. Digital Resource Management Expansion

   The Digital Library Officer module can be extended to manage e-books, research papers, multimedia resources, and online journals in addition to physical books.

Implementing these recommendations will make the system more intelligent, scalable, secure, and aligned with future technological advancements and user requirements. It is also recommended that Mekdela Amba University adopt and maintain the system for practical use within the university library environment.

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